

AI-Based Landmark-Driven Muscle Hypertrophy Tracking System Using 2D Computer Vision

B.Tech.

**Project Work in
Department of CSE**



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Certificate

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AI-Based Landmark-Driven Muscle Hypertrophy Tracking System Using 2D Computer Vision

The project work is submitted in partial fulfilment of academic requirements for the award of B. Tech. Degree in the Department of Computer Science Engineering of the Faculty of Engineering and Technology of M. S. Ramaiah University of Applied Sciences. The project report submitted herewith is a result of our own work and in conformance to the guidelines on plagiarism as laid out in the University Student Handbook. All sections of the text and results which have been obtained from other sources are fully referenced. We understand that cheating and plagiarism constitute a breach of University regulations, hence this project report has been passed through plagiarism check and the report has been submitted to the supervisor.

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Abstract

The primary goal of this project involves developing an AI-based muscle hypertrophy tracking system that automatically measures body dimensions and monitors muscle growth over time. Fitness enthusiasts and individuals engaged in strength training often face challenges in accurately tracking their physical progress, as traditional methods rely on manual measurements and subjective visual comparisons, which lack consistency and precision. With the rapid advancement of artificial intelligence and computer vision, there is an increasing demand for intelligent systems that can support objective and data-driven fitness analysis. The proposed system utilizes computer vision techniques to extract anatomical landmarks from real-time images and convert them into meaningful body measurements. The process begins with capturing multi-view images (front and side), followed by pose estimation using frameworks such as MediaPipe Pose to detect key body points. These landmarks are then processed to compute measurements such as arm length, shoulder width, and thigh length. A scaling mechanism is applied to convert pixel-based values into real-world units, ensuring accurate and consistent results. A key feature of the system is the use of multi-view geometric modeling, where width and thickness information from different perspectives are combined to approximate muscle size. This is achieved through a pseudo-3D approach, where muscle cross-sections are modeled using an elliptical formulation. The system also incorporates validation mechanisms to ensure correct posture during capture, thereby improving measurement reliability. The experimental results demonstrate that the system produces consistent measurements with low error margins and effectively tracks muscle growth trends over time. The solution operates efficiently using standard hardware and provides repeatable results, making it suitable for practical fitness applications as well as academic purposes.

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Nomenclature

<i>AI</i>	<i>Artificial Intelligence</i>
<i>API</i>	<i>Application Programming Interface</i>
<i>BPM</i>	<i>Beats Per Minute</i>
<i>CPU</i>	<i>Central Processing Unit</i>
<i>DL</i>	<i>Deep Learning</i>
<i>Epoch</i>	<i>One complete training cycle of the model</i>
<i>FFT</i>	<i>Fast Fourier Transform</i>
<i>GPU</i>	<i>Graphics Processing Unit</i>
<i>LR</i>	<i>Learning Rate</i>
<i>LSTM</i>	<i>Long Short-Term Memory</i>
<i>MIDI</i>	<i>Musical Instrument Digital Interface</i>
<i>ML</i>	<i>Machine Learning</i>
<i>Seq</i>	<i>Sequence Length (number of notes)</i>
<i>Token</i>	<i>Numerical representation of a musical note</i>
<i>UI</i>	<i>User Interface</i>

Abbreviation and Acronyms

AI	Artificial Intelligence
API	Application Programming Interface
BPM	Beats Per Minute
CPU	Central Processing Unit
DL	Deep Learning
FFT	Fast Fourier Transform
GPU	Graphics Processing Unit
LSTM	Long Short-Term Memory
MIDI	Musical Instrument Digital Interface
ML	Machine Learning
RNN	Recurrent Neural Network
Seq2Seq	Sequence-to-Sequence Model
UI	User Interface

1. Introduction

Preamble to the Chapter:

The growing intersection between artificial intelligence and human-focused applications have provided revolutionary prospects in several fields, such as healthcare, sports science, and fitness technology. This chapter provides the background in the creation of a landmark-based AI-based muscle hypertrophy tracking system based on 2D computer vision. It starts with placing the problem in the larger context of digital anthropometry and performance tracking and critically analyzes the inadequacies of the traditional methods of muscle measurement. The chapter also explains the purpose of automating the tracking of hypertrophy and outlines the conceptual and technical framework of the proposed system. In such a way, it preconditions the knowledge of how the current computational methods can be used to substitute the subjective and error-prone methods with the data-driven and scalable and accurate solutions.

Traditional fitness ecosystems have progress tracking that can be informal, inconsistent, and extremely reliant on user discipline. Although people put in much time and effort in strength training, the systems that can be used to measure physical changes are crude. This lack of connection between work and quantifiable feedback is a point of major deficiency which can be bridged by smart systems. The proposed solution seeks to fill this gap by taking advantage of recent developments in computer vision, pose estimation and geometric modeling to provide an automated pipeline of hypertrophy tracking, which is both accessible and reliable. This chapter, hence, not only lays out the technical field, but puts in a broader context the implications of such a system in improving the user experience, the scientific precision and the overall fitness results.

1.1 Introduction

The fast development of artificial intelligence, especially in the domain of computer vision, has facilitated the precision with which machines process and analyze visual data to a degree that was not possible before. Computer vision systems can now detect objects, patterns and interpret spatial relationships in images and videos. These features have been effectively implemented in various fields including medical imaging, autonomous driving, surveillance and augmented reality. One of such applications that are especially pertinent to this technological advancement is the interpretation of the human body in terms of visual data, which is the foundation of digital anthropometry.

Digital anthropometry is the process of quantifying and analyzing the human body with digital devices, often by processing images and employing computational models. Digital methods can be used to obtain body dimensions non-invasively through cameras and algorithms, unlike the traditional technique of anthropometry that involves the use of physical tools like measuring tapes and calipers. The benefits of this change of manual to automated measuring include greater accuracy, repeatability, and scalability. In addition, digital systems have the capability of storing and analyzing large amounts of data over time, allowing longitudinal examination of physical alterations that would be hard to accomplish using manual procedures.

When it comes to fitness and strength training, muscle hypertrophy, or the expansion of the muscles due to resistance training, is one of the most critical measures of the progress. One of the aims of athletes, bodybuilders, and fitness enthusiasts is hypertrophy, which is directly proportional to strength, beauty, and physical overall performance. Proper monitoring of muscle growth is thus critical to assess training program efficacy, make informed modification and to keep motivation. Nevertheless, even though it is a significant aspect, the devices used to monitor hypertrophy are still mostly insufficient.

Mostly muscle growth is measured using a measuring tape that measures the circumference of certain body parts like arms, chest, waist and thighs. This is a cheap and easy method, but very prone to human error. Inconsistencies in measurement, including the difference in the tension of the tape, positioning and posture, may cause unequal

outcomes. Also, manual measures are normally taken at uneven times, and thus it is hard to have a continuous and dependable record of the progress. These drawbacks make the use of tape measures inaccurate when it comes to measurement of precise hypertrophy.

The next popular method is subjective visual comparison where people evaluate their progress by looking at themselves in the mirror or comparing photos of themselves at various times. Though this approach may offer some overall idea of change, it is not objective and has no measurable indicators. Lighting conditions, camera angles, and user bias are some of the factors that may affect the perception considerably and cause incorrect conclusions. Moreover, visual comparisons lack organized information that could be analyzed or utilized to make training decisions.

Periodic photo comparison is an attempt to bring in some sort of consistency since pictures are taken under controlled conditions. Users can record progress photos at a certain rate, with a similar pose, lighting and background. Although this method is more organized than visual comparison in a casual manner, it is also based on manual interpretation and does not have automated ways of measuring. Consequently, it is unable to give accurate or practical information on muscle development.

These constraints underscore a key issue with fitness tracking, which is that there are no objective, automated and scalable systems that measure hypertrophy. With the adoption of technology in the fitness industry, there is an increasing need to have solutions that can address these issues and deliver accurate and data-driven information. The increased influence of individualized fitness, online coaching, and remote health monitoring only enhances this need since all these necessitate the availability and reliability of accurate data in order to be effective.

The suggested system will fulfill this requirement by using the potential of 2D computer vision and artificial intelligence to develop an automated hypertrophy tracking system. The essence of this system lies in the application of pose estimation algorithms, which are able to detect important anatomical landmarks on the human body by using pictures. Such features as joints and body shapes can be used as points of reference to determine

distances and proportions. The system can approximate body sizes and monitor variations over time by examining the spatial relations of these landmarks.

Geometric modeling to approximate muscle volume and growth is one of the main innovations of the proposed approach. The system uses a pseudo-3D approach to estimate volumetric changes based on planar measurements because 2D images do not directly give information on depth. This includes geometric modeling of body parts (cylinders or ellipsoids) according to landmark-based dimensions. The system is able to estimate muscle size changes with a reasonable level of accuracy by comparing these representations at various time points.

The combined pose estimation and geometric modeling allows the system to automate all the hypertrophy tracking process. The users are required to take a shot of their body with a normal camera, say a smartphone. These images are then fed into the system to identify landmarks, determine measurements and produce insights. This eradicates the use of manual measurements and the possibility of human error. Moreover, the system is automated which guarantees the consistency of measurements, so the progress can be followed more precisely.

Scalability is another feature of the proposed system. In contrast to the previous approach, where it was necessary to interact with a person and work manually, the AI-based one can be implemented on a large scale and be used on various platforms, such as web-based interfaces and mobile applications. This renders it usable by many users, irrespective of their location and the degree of expertise. Moreover, the system is able to store and analyze the data in the long-term, which allows users to visualize their progress and see trends in their training performances.

The reason why this system has been developed is not just to enhance the accuracy in measurements but also to enhance the overall fitness experience. The system can be used to inform users with objective and actionable information about their training routines to make informed decisions. Indicatively, users are able to determine which muscle groups are developing well and which need more attention. This amount of feedback can greatly enhance the effectiveness and results of training.

Additionally, the system can be used in other applications that are not necessarily related to individual fitness. Coaches and trainers can also use it to track the progress of their clients, allowing them to use more personalized and data-driven coaching plans. It can be applied in clinical practice to help in rehabilitation and physical therapy as it can be used to monitor muscle growth and healing. Non-invasive and remote measurement of body measurements also creates the opportunities of telemedicine and remote health monitoring.

Although such a system has its benefits, there are a few challenges associated with its development. The correctness of landmark detection is one of the major challenges, particularly when the conditions are dissimilar like when the light is different, the clothing, and body pose. The other issue is the fact that approximation of 3D information based on the use of 2D images entails assumptions and therefore errors. The solution to these challenges is in designing an algorithm with care, preprocessing of data, and ground truth measurements.

To conclude, the advent of the AI-based landmark-driven muscle hypertrophy tracking system is an important development in the area of fitness technology. The proposed system will provide an innovative solution to the constraints of the traditional measurement techniques by combining computer vision, pose estimation, and geometric modeling. It offers a system of automated, objective and scalable tracking of hypertrophy, which can revolutionize the way people and professionals keep track of physical growth. The chapter has described the background, intent and the essence of the system which will prepare the ground for a further discussion of its design, implementation and analysis in the following chapters.

1.2 Limitations of Existing Systems

Although the development of the current state of the art in the field of fitness technology has been positive, the existing ecosystem of muscle measurement and body tracking systems continues to be essentially constrained with regards to providing accurate, consistent, and longitudinal information on the issue of muscle hypertrophy. Although contemporary devices feature the aspects of digital tracking, wearable integration, and mobile apps, they might not serve the main need of precisely quantifying muscle growth over time. The reason behind this gap is due to a combination of methodological incompetence, a lack of standardization and poor integration of the latest computational methods, including computer vision and geometric modeling. Consequently, both users and professionals still have to use the imperfect systems that do not reflect the intricacy of human body development comprehensively. The section is a critical analysis of these limitations classifying them into manual measurement techniques, visual comparison techniques and the available digital systems.

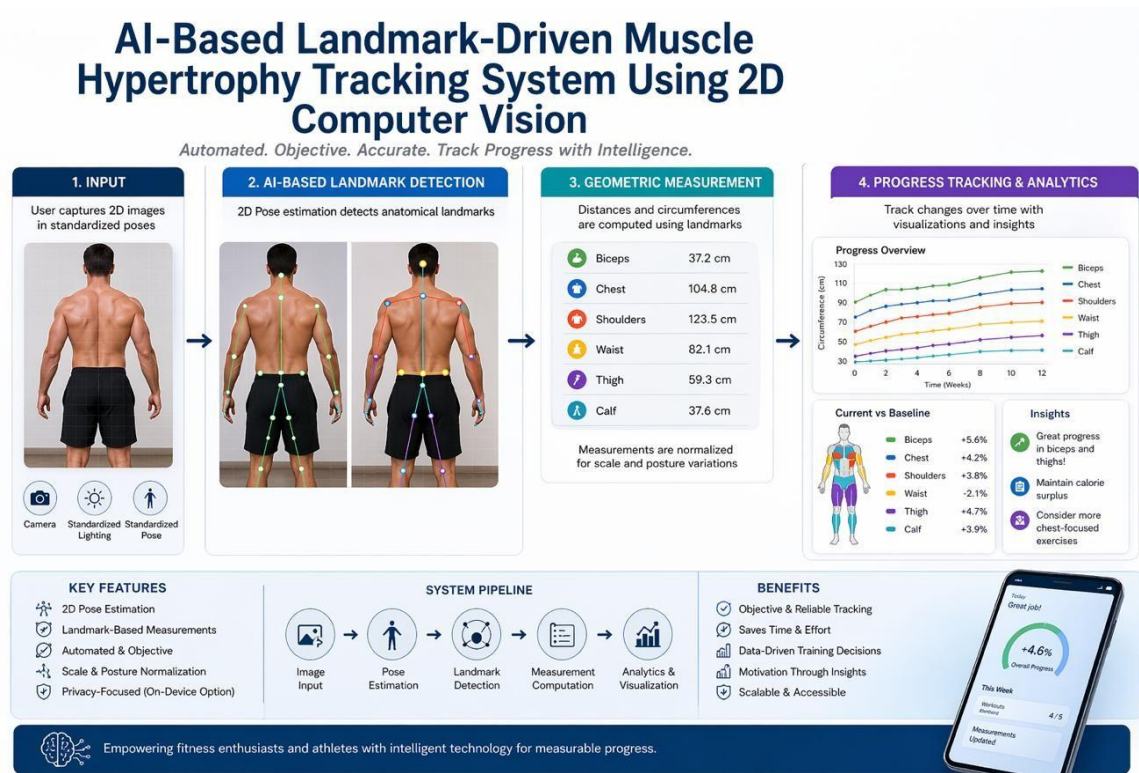


Figure 1. 1 AI Based hypertrophy tracking system overview

1.2.1 Manual Measurement Limitations

The conventional manual methods of measurement, especially the application of measuring tapes has long been the standard method of measuring the growth of the muscle. These techniques are quite popular because they are simple, cheap and do not need much technology. Nevertheless, even though they are easy to measure, manual measurements have a few inherent limitations that seriously compromise their reliability and validity, particularly when monitoring long-term hypertrophy.

Human error is one of the most significant problems related to manual measurements. Tape measurements need to be made with precision in the placement of the tape, tension and alignment of the tape around the part of the body that is being measured. Any slight variation in any of these factors can cause great differences in the recorded values. As an example, a small change in the location of the tape on the arm or thigh may result in varying circumference measurements, and it may be challenging to decide whether the changes were actually the result of muscle increase or measurement errors. This variation adds noise into the data and makes it less useful in analysis.

Besides, human error, manual measurements are inconsistent among sessions. Repeatability of measurement conditions is difficult to achieve because a variety of factors can affect the results: posture, muscle contraction state, hydration level, time of the day and so on. An example is when a muscle is measured in flexion and in a relaxed position can have dramatically different values. On the same note, difference in body posture or body position can change the perceived circumference of a body part. In the absence of rigorous standardization procedures, it is almost impossible to determine that measurements made at other times are directly comparable.

The reliance on measurement technique is another major weakness. Some degree of skill and experience is needed to do the measurement properly, especially in locating anatomical landmarks, and in maintaining a consistent location of the tape. In most instances, self-measurements are taken without professional direction thus resulting in additional inaccuracies. Inter-observer variability may arise even when the measurements are carried out by trained people with the variation being in the results given when the

same situation is carried out by different individuals. This is an absence of consistency and thus the validity of the information is undermined and the scope of its use is restricted both personally and professionally.

Manual measurement systems pose a huge challenge in the form of standardization. Although there are standards regarding the measurement of certain parts of the body, it is hard to apply such norms in practice. Users usually do not follow recommended procedures because of convenience, ignorance or environmental factors. This leaves the gathered data devoid of uniformity and therefore, inappropriate to perform accurate longitudinal analysis. In the long-run, these discrepancies tend to be added up, resulting in inaccurate trends and even false conclusions on muscle growth.

Moreover, the manual approaches cannot be digitally scaled. Each of the measurements needs active human intervention, thus it becomes time-consuming and cannot be practiced on a regular basis. This also implies that data cannot be automated and they have to be recorded and handled manually, exposing the possibility of mistakes and decreasing effectiveness. Manual measurement is not up to the expectations in the context of the modern fitness applications where users demand seamless and real-time feedback.

To conclude, although manual measurement methods offer a simple method of monitoring muscle size, their error susceptibility, non-reliability, reliance on methods and inability to be standardized severely hinder their usefulness. These disadvantages point out the necessity of automated methods capable of providing precise and consistent measurements without the interference of human operators.

1.2.2 Visual Comparison Limitations

Another technique which is widely employed to measure the growth of the muscle is the visual comparison method, which is mostly utilized by the fitness enthusiasts and amateur practitioners. This technique normally includes the comparison of body images of the body at various time intervals, commonly known as before-and-after photos. Although it is an intuitive, simple method, in essence, this method is constrained by its subjectivity and the absence of quantitative rigor.

Lack of quantifiable data is one of the most significant drawbacks of the visual comparison. In contrast to the numerical measurement, where statistical analysis is possible and can be monitored over time, visual assessment is all about perception. This complicates the process of measuring the level of muscle growth or even identifying any minor changes which might not be noticeable instantly. In the absence of objective metrics, users are expected to make judgements based on their own feelings which are subject to cognitive biases and external influences.

The aspect of lighting conditions is important to the perception of muscle definition and size in an image. The way the lighting is varied in intensity, direction and color can change the appearance of the body dramatically. As an example, directional lighting may develop shadows that can make the muscles more defined, creating an illusion of increased hypertrophy. On the other hand, flat or diffused lighting may hide the definition of the muscles leaving the body to look less developed. These discrepancies complicate the comparisons of images that have been taken in varying circumstances because what seems like changes maybe as a result of light and not necessarily real physical development.

The angle and the placement of the cameras make visual comparisons even more complicated. The few variations in camera height, distance or orientation have the ability to alter proportions and influence the appearance of body parts in the picture. An example is that a closer camera shot can make some of the muscles look bigger, and a higher angle shot can make the body look leaner. Even small changes in pose/ posture can cause a big difference in perceived muscle size. Unless these variables are tightly controlled, visual comparisons cannot be as consistent as they need to be tracked accurately.

The other major problem is the impact of user bias. People tend to have pre-existing assumptions or emotional stakes in their development and it may influence the way they perceive visual information. It may result in either overestimating or underestimating growth of the muscles, depending on the thinking of the individual. Confirmation bias can result in users thinking that there is an improvement where none, whereas negative self

perceptions can result in the reverse. These psychological aspects further decrease the accuracy of the visual comparison as a measuring device.

Additionally, visual comparison techniques cannot be used to track accurately or in granules. Although they can be employed in the observation of large scale changes over long periods of time, they cannot be employed to detect incremental progress. The failure to resolve subtle changes is a major drawback in the context of hypertrophy where the changes may be subtle and gradual. Such imprecision makes it hard to judge the efficiency of particular training interventions or to maximize workout programs on the basis of quantifiable results.

Moreover, visual techniques are not conducive to structured data storage and analysis. Photos can be placed into galleries or apps, yet without metadata attached or metadata extracted (through measurement), they will be unorganized and hardly used. This restricts the capacity to make longitudinal analysis, see trends or incorporate data into larger fitness tracking frameworks.

On the whole, although visual comparison is the easiest and convenient method of tracking physical changes, its subjectivity, vulnerability to external influences, and non-quantitative data predisposes it to inaccurate hypertrophy monitoring. These restrictions emphasize the necessity of the systems that would be able to transform visual data into objectively structured measurements.

1.2.3 Existing Digital Systems Limitations

With the proliferation of digital health and fitness technologies, numerous applications and devices have been developed to assist users in tracking their physical attributes. These systems often incorporate features such as body composition analysis, activity tracking, and health monitoring. While they represent a significant advancement over traditional methods, they still fall short in addressing the specific requirements of muscle hypertrophy tracking.

Most existing digital systems focus on generalized health metrics such as body fat percentage, body mass index (BMI), and overall weight. While these metrics provide

useful insights into general health and fitness, they do not offer detailed information about muscle growth. For example, BMI is a simple ratio of weight to height and does not differentiate between muscle and fat mass. As a result, it is not a reliable indicator of hypertrophy. Similarly, body fat estimation techniques, whether based on bioelectrical impedance or visual analysis, provide only a partial view of body composition and do not directly measure muscle size.

Another limitation of current digital systems is their emphasis on static measurements. Many applications allow users to input or record body measurements at specific points in time, but they do not provide robust mechanisms for tracking changes longitudinally. The lack of continuous data collection and analysis limits the ability to observe trends and patterns in muscle growth. Without longitudinal tracking, users cannot effectively evaluate the impact of their training programs over time.

Furthermore, existing systems often lack multi-view geometric modeling capabilities. Most digital tools rely on single-view images or simple input data, which restricts their ability to accurately represent the three-dimensional structure of the human body. Without multi-view analysis, it is difficult to capture the full geometry of muscle groups, leading to incomplete or inaccurate measurements. This is particularly problematic for muscles that have complex shapes and orientations, which cannot be adequately represented in a single 2D perspective.

Measurement validation is another area where current systems fall short. Many applications do not provide mechanisms to verify the accuracy of their measurements or to calibrate their models against real-world data. This lack of validation raises concerns about the reliability of the results, especially when used for critical decision-making. In the absence of ground truth comparison or error estimation, users have limited confidence in the data generated by these systems.

Additionally, most digital fitness tools do not offer muscle-specific analysis. They tend to treat the body as a whole or focus on broad regions, rather than isolating individual muscle groups. This limits their usefulness for targeted training and performance optimization. For example, a user may want to track the growth of specific muscles such

as the biceps, quadriceps, or deltoids, but existing systems do not provide the granularity required for such analysis.

Another important limitation is the lack of integration between different data sources and analytical models. Many systems operate in isolation, without leveraging advanced techniques such as pose estimation, landmark detection, or geometric modeling. This results in fragmented functionality and prevents the development of comprehensive tracking solutions. The absence of a unified framework that combines visual data processing with measurement extraction and analysis is a key gap in current technologies.

In conclusion, while existing digital systems represent a step forward in fitness tracking, they are not equipped to handle the complexities of muscle hypertrophy analysis. Their focus on generalized metrics, lack of longitudinal tracking, absence of multi-view modeling, limited validation, and insufficient muscle-specific analysis highlight the need for more advanced and integrated solutions. The proposed AI-based system seeks to address these shortcomings by providing a comprehensive framework for automated, accurate, and scalable hypertrophy tracking using 2D computer vision.

1.3 Research Motivation

Research Focus and Motivation

The current project is motivated by the increasing demand for addressing the challenges of conventional approaches to track muscle volume, and to develop a structured, quantitative approach to muscle hypertrophy study. In the field of fitness training and strength training, the accurate measurement of growing muscles plays a crucial role in assessing progress, improving training programs, as well as keeping progression of body growth consistent. But the current methods are primarily based on manual assessment and qualitative observation, and are generally not accurate or consistent.

The recent progress in artificial intelligence and computer vision technologies offers an opportunity to automate the measurement and tracking of body growth. These systems can process visual information, identify patterns, and interpret information in an

automated manner. The proposed project building on these technologies, design an intelligent system that can measure body sizes and track hypertrophy.

The research interest is then focused on combining pose estimation, geometric modelling and data analysis. This not only enhances measurement accuracy, but also allows for scalable measurement of hypertrophy. This system converts visual information into quantitative data, making data-driven analysis in fitness possible.

Key Challenges Identified

A major issue in current systems is the absence of automated measurement systems. Conventional techniques rely heavily on manual measurements using tapes, leading to inconsistencies and errors. Variations in measurement techniques can result in unreliable measurements. This inconsistency hampers longitudinal measurements.

A key limitation of existing systems is the lack of depth estimation. The majority of image-based measurement methods use a single view from a camera, which only provides 2-dimensional data. This introduces a lack of three-dimensional detail of the body in question. This and also impacts on the accuracy of measurement, especially for estimating muscle size where volume changes occur. Lack of a formal system of longitudinal monitoring is another major problem. Current systems tend to offer "snapshot" measurements without historical data storage. Without a framework for storing and analysing measurements, it's hard to track progress or get an idea of trends in data. Furthermore, manual measurements are highly variable due to variations in posture, measurement techniques and environment. Minimal changes in posture or placement can result in large differences. This affects the data's accuracy and its capacity to be tracked over time.

Research Focus

Our project aims to tackle these issues using a multi-front computer-vision approach. A primary focus is on the extraction of anatomical landmarks using pose estimation. These techniques are able to extract landmarks on the body like joints and pose references from images or video. These key points offer a robust basis for measurement extraction, with all calculations being properly normalized.

We also present a multi-view measurement system. The system captures the body from both the front and side to gain additional information about the body's structure. The front view allows for width measurements to be obtained, whereas the side view allows for thickness measurements to be obtained. This approach overcomes the limitations of single-view 3D systems and allows a better representation of the body.

The project also places an emphasis on pseudo-3D geometric modelling. As direct 3D measurement is not feasible with 2D cameras, geometric models are used to approximate muscle volume. The system uses simplifications of muscle cross-sections, like ellipses, to approximate cross-sectional area and changes in volume. This offers a better depiction of hypertrophy than simple linear changes. Second, it's important to support tracking growth over time. The system should retain the data over multiple measurements in a structured manner, enabling the user to interpret the data over time. This enables the system to track changes over time, measure growth, and offer insights into muscle development. This enables the system to evolve from a measurement tool into an analysis system.

Motivation

The main driving force behind this project is the desire to develop a system to minimize manual measurements. Manual methods can be tedious, susceptible to mistakes and inaccurate. The automated measurement system ensures that the measurements are objective, consistent and user-independent. The other major driving factor is that of consistency. This is essential in order to monitor progress over time, as inconsistencies in measurement may skew claims of improvement. The pose estimation and consistent measurement across positional landmarks ensures consistency.

The project also seeks to allow quantitative fitness tracking. Users can rely on data rather than their subjective assessments. This will enable more objective judgment of all training sessions and will help plan your workout. Structured data also allows for data to be visualised and analysed, so progress can be more readily understood.

Accessibility is another important motivation. The system can be implemented using inexpensive equipment - such as a webcam - which ensures accessibility. Unlike complex 3D scanning systems, the proposed system can be used with widely available technology. This allows automated muscle tracking methods to be performed without the need for special equipment.

1.4 Core Idea of the Proposed System

The suggested AI-muscle hypertrophy tracking system with landmarks is essentially aimed at closing the gap between subjective and objective, data-driven fitness analysis. The system is essentially a combination of multi-view 2D computer vision and geometric modeling to measure muscle size and track its development with time. Compared to the traditional methods, which are based on either manual measurements or single visual comparisons, this system presents an organized computational pathway that converts raw image data into substantial quantitative information. The point is not just to take pictures but to process them with sophisticated algorithms that can in turn determine body sizes and estimate volumetric remodelling, thus allowing the robust and scalable tracking of hypertrophy.

One of the principles of the system is the adoption of multi-view pose estimation which is the utilization of front and side view of the human body. The human anatomy is naturally three dimensional and a 2D view with a single perspective creates great limitations especially in depicting depth and thickness. The system can collect orthogonal views on the body parts, usually frontal and lateral, and through this the system can extract complementary information on body dimensions. The front view mainly gives information about the width like shoulder span, arm girth, and torso breadth and the side view gives the thickness like chest depth and limb profile.

The key to this process is the use of pose estimation algorithms, whose task is to detect and localize anatomic landmarks on a human body. These are landmarks containing important joints and reference points like shoulders, elbows, hips, knees and other important points used to mark the skeletal structure. Even the more challenging conditions of lighting, clothes, and background can be successfully detected by modern pose estimation models, which are typically driven by deep learning systems. These landmarks are used as measuring anchors once identified. The body dimensions are calculated through the spatial relationships that exist between the landmarks, including the distance, angles, and relative proportions, to create a consistent and repeatable method of calculating the body dimensions.

The translating of these landmark coordinates into physical measurements that are meaningful and that can be derived using geometric formulas is the next stage of the system. The input data is obtained on the basis of 2D images, and as such, the system should consider scale and perspective to guarantee that the values that are computed are reflective of real-world dimensions. This is normally obtained by use of normalization methods, calibration standards or proportion scaling that is based on known body ratios. Using the landmark data and applying geometric principles, the system can estimate lengths, widths and circumferences of the different body parts. As an illustration, the upper arm length can be estimated by the distance between landmarks of the shoulders and the elbow and the width of the muscles can be estimated by the horizontal displacement of the landmarks in the front view.

An approximation of muscle cross-sectional area with an elliptical model is one of the most innovative features of the proposed system. Muscle shapes are complicated and irregular, but to simplify the computation, they may be modeled using simplified geometries. The ellipse is a useful and mathematically solvable model to describe the cross-section of most muscle groups. Using these measurements the system identifies the major and minor axes of an ellipse with respect to a particular muscle group by combining the width measurements in the front view with the thickness measurements in the side view. The area of the cross-section can then be estimated by using standard geometric formulae which gives an approximation of the size of the muscle. This will enable the

system to go beyond mere linear measures and integrate a volumetric view, which is more in keeping with the hypertrophy concept.

With time, the system stores the measurements as users take the pictures at various stages of training. This allows the development of a long-term dataset that would capture the muscle development. The system will be able to detect trends i.e. steady growth, plateau or regressions by examining change in computed areas and dimensions. Intuitive graphs and dashboards then visualize these trends and enable the users to monitor their progress in an actionable and easy to understand way. There is also the capacity to store historical data, which makes it easier to provide a comparative analysis and, thus, to gauge the effectiveness of various training programs or interventions.

Automation is another aspect of the main concept. The whole pipeline, which includes image capture, landmark detection, measurement computation, and trend visualization, is created to be run with few user interactions. This not only minimizes the chances of human error, it also increases usability and accessibility. Users do not need to possess technical skills or observe complicated measurement procedures; they are engaged with a system which can smoothly translate visual input into a structured output. This degree of automation is critical in being able to maintain consistency in sessions, which is important in long-term tracking.

Moreover, the system focuses on scalability and adaptability. It does not require special equipment as it uses the conventional 2D imaging devices like smartphone cameras, which can be used by a wide range of users. The underlying algorithms can be implemented on a multiple range of platforms such as mobile applications and web interfaces to allow mass adoption. Moreover, the system is designed in a way that makes it easy to add or upgrade in the future, so that more sophisticated models, better calibration methods or other sources of data can be added.

Essentially, the proposed technique will mark a paradigm shift in the measurement and analysis of muscle hypertrophy. It substitutes subjective decision-making and human labor with a strict system of calculations that capitalizes on the advantages of artificial intelligence and geometrical thinking. The system uses a combination of multi-view pose

estimation and elliptical modeling and longitudinal data analysis to offer a comprehensive solution to muscle growth tracking in a precise, consistent, and scalable way. The transformation is not only more accurate in measurement but also provides the users with more knowledge about their physical progress, which leads to more productive and informed fitness.

1.5 Contribution of the Project

The given AI-based muscle hypertrophy-monitoring system based on landmarks offers a bundle of contributions that in combination contribute to the development of the fields of fitness analytics, digital anthropometry, and applied computer vision. These contributions are not just the additions to current systems but are a systematic change to automated, data-driven and scientifically based approaches to monitoring muscle growth. The system overcomes the long-standing issues of accuracy, consistency, and scalability of hypertrophy measurement by introducing pose estimation, multi-view analysis and geometric modeling into a single framework.

The development of a landmark-driven measurement framework is one of the major contributions of this project. The conventional measurement systems are very manualistic, as the users or trainers physically locate measurement points on the body. This is a process by itself subject to variability because of the difference in technique, positioning and interpretation. Conversely, the proposed system uses high-quality pose estimation algorithms to automatically identify body anatomical landmarks like joints and relevant locations on the body. Such landmarks are used to provide a standard anchor point on which all later measurements can be pegged. The system also removes subjective variability by basing the measurement process on consistent and algorithmically determined reference points, which also guarantees repeatability across sessions. This is not only done to enhance accuracy in the measurements, but also to provide a strong base on which to carry out further computational analysis.

A second important input is the addition of multi-view geometric modeling in order to increase the accuracy of measurements. Human body is a 3 dimensional form and recording its geometry in a 2D image will inherently lead to loss of information and

especially depth. The proposed system surpasses this shortcoming by integrating front and side views of the subject matter. The front view will give the width and side view will give the depth and thickness. The combination of these two viewpoints will re-assemble a more complete picture of body geometry, a three dimensional structure, which is roughly approximated by two dimensional inputs. This multi-view method is much better at fidelity of measurements and makes it possible to estimate muscle size more accurately. pseudo-3D muscle size estimation is a new and significant addition to the work of this project. Although in a real-world scenario, true 3D reconstruction would need specific hardware devices, including depth sensors or multi-camera systems, the offered system is able to deliver the same with the help of computation modeling. The system converts 2D measurements into volumetric approximations by approximating the shape of muscle groups with geometric primitives (ellipses and cylinders). The cross-section of a muscle can be used as an example, where the width and the thickness of the cross-section of a muscle are measured on various views, and the major and minor axes of an elliptical shape are obtained. It will enable the system to estimate the cross-sectional area and derive changes in muscle volume with time. This method gives a more valuable representation of hypertrophy than linear measurements, and is much closer to physiological definitions of muscle growth.

The other cornerstone contribution of the system is longitudinal tracking of muscle growth. As opposed to traditional approaches, which offer single-point body measurements, the proposed framework will be able to record and save the information at various time points. This allows building of a temporal dataset indicating muscle development progression. Through the analysis of this data, the system is able to determine trends, including constant growth, stagnation, or regression, and provide these insights in a visual format in the form of graphs and charts. This is what makes the tracking process an ongoing analytical process and enables users to track their progress with a lot of accuracy. It also allows making evidence-based decisions, and users and trainers can modify training programs in response to observed outcomes.

The other significant contribution is the focus on validation-based measurement reliability. The absence of mechanisms to verify the accuracy of the outputs of their digital fitness tools is one of the greatest failures of the available ones. The proposed system

countermeasures this problem by introducing validation techniques into the measurement pipeline. These can involve cross-referencing the measurements made in other views, normalization and calibration processes and consistency in repeated measurements. The project improves the credibility and reliability of the data generated by the system by embedding the validation in the system. It is especially significant in the areas where, precise measurement is crucial like professional training, rehabilitation and research. The system also helps to enhance better accessibility and scalability in hypertrophy tracking. It is available to a large number of users since it does not require any specific tools, just a normal imaging tool like a smartphone camera. The computational framework is applicable to diverse systems, such as mobile applications and web-based systems, to ensure it can easily integrate with an existing fitness ecosystem. This is so scalable, so that the advantages of sophisticated measurement methods need not be restricted to specialized settings but can be applied to ordinary users.

Besides technical contributions, the project also presents some paradigm shift in terms of interaction of the user with fitness data. Conventional systems are usually manual and need to be interpreted manually, and this puts a cognitive load on the user. The suggested system automatizes the whole procedure, including the process of data collection, analysis and visualization. This does not only enhance efficiency but also the level of user interaction since it can offer relevant and practical insights. The objective feedback allows users to monitor their progress and easily identify areas of improvement and keep themselves motivated.

Moreover, the system is modular which will enable further expansion in the future. Existing framework can be expanded with new algorithms, better models or other data sources without breaking the existing framework. This is to make sure the system is able to keep up with any changes in technology and will be relevant and effective in the long run. Possible future directions are to be integrated with wearable devices, to include machine learning models to predictive analysis, or to refine geometric modeling techniques to achieve better accuracy.

In general, the impact of this project is not limited to the direct use of muscle hypertrophy monitoring. They show the possibility of artificial intelligence, computer vision, and geometric modeling to address complex problems in the real world. The project innovates by improving the shortcomings of the existing systems and coming up with new solutions based on accuracy, automation, and scalability in the field of fitness analytics.

1.6 Organization of the Report

The outline of this report is well-developed to offer a logical and thorough flow of ideas starting with the basics, to details of implementation and conclusive results. One chapter is a continuation of the other so the reader of the article is guaranteed to get a clear picture about both the theoretical and practical side of the proposed system. The structure demonstrates the organized way of handling the problem-solving process, consisting of context and analysis, design and implementation, and evaluation and future directions.

Chapter 2 is concerned with the problem formulation and literature review. This chapter is a comprehensive review of the available studies on topics like computer vision, pose estimation, digital anthropometry, and fitness tracking systems. It analyses the methodologies, strengths, and weaknesses of the existing approaches and what is missing that the proposed system seeks to fill. The chapter draws upon the work of other scholars to form a solid theoretical framework and explains why the suggested solution is necessary. It also narrows down the problem space through the clear definition of the challenges that are encountered with the accurate and automated tracking of hypertrophy.

Chapter 3 is a formal and structured statement of the problem. Based on the results of the literature review, this chapter will present the precise problems that the project will address. It also establishes the scope of the system, its goals and criteria of success. The problem statement is a crucial bridge between theoretical knowledge and practice, which makes the system design correlated with the clearly established objectives.

Chapter 4 elaborates the system architecture and methodology that are the center of the project. This chapter outlines the general structure of the system, its components, data flow, and computations. It describes the process by which input images are manipulated by pose estimation algorithms to obtain landmarks, the use of landmarks to compute measurements, and the use of geometric modeling to approximate muscle size. Implementation details, including preprocessing steps, model selection, calibration methods, and data storage mechanisms are also discussed in the chapter. This chapter allows one to replicate the work by offering a detailed description of the methodology and understand the work of the system better. Chapter 5 introduces results and discussion where performance of the system is analyzed with the help of the experimental data. Quantitative measurements accuracy, comparison of traditional and existing digital methods, and visualization of hypertrophy tracking results are all included in this chapter.

The report is concluded in chapter 6 and shows future scope. This chapter summarizes the most important contributions and findings of the project and speculates on its importance within the framework of fitness technology and computer vision. It also discusses possible future research and development directions that may be undertaken including improving the precision of pseudo-3D modeling, real-time capabilities, or extending the system to accommodate other health and fitness indicators. The conclusion supports the worth of the project and underlines its possibility to be further innovated.

The report can also contain other parts besides these key chapters, like appendices, references and acknowledgements. These parts give more background information, specifics, and references to substantiate the main information. Clarity, coherence, and completeness are guaranteed by the overall organization that helps the reader to navigate the report successfully and have a comprehensive view of the proposed system.

In this organized manner, the report will not only record how the AI-based hypertrophy tracking system developed but will also have a contribution to the overall field of knowledge. It is a resource of technical knowledge and a source of further activity, showing how progressive methods of computation are used to find solutions to real-life challenges in the field of fitness and health monitoring.

2. Literature Review and Problem Formulation

Preamble to the Chapter

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Chapter 5 introduces results and discussion where performance of the system is analyzed with the help of the experimental data. Quantitative measurements accuracy, comparison of traditional and existing digital methods, and visualization of hypertrophy tracking results are all included in this chapter. It also talks about the consequences of the findings, the strengths, and weaknesses of the system. The discussion offers acute reflections on the efficiency of the suggested solution and suggests areas to improve.

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2.1 Background Theory

2.1.1 Digital Anthropometry

The traditional anthropometric approach was very dependent on the use of physical tools (measuring tapes, calipers, skilled practitioners) and the body measurement was difficult to be performed on a large scale or repeated because of the time and risk of human error. In response to these limitations, there was a response to the field of digital anthropometry that used computational tools to automatically extract body measurements from images or sensor data. Digital anthropometry, which integrates photogrammetry, depth sensing, and machine learning, provides contactless, consistent and scalable measurement workflows across industries, including healthcare, sports science, ergonomics and apparel.

Digital anthropometry forms the base layer of measurement in the field of hypertrophy tracking. As opposed to the individual gym goer's feeling of progress, digital techniques enable objective circumference and volume estimates to be determined from the skeletal landmarks identified in images. The difficulty is controlling the variables of measurement in a room with natural lighting, occlusions by clothing, camera angle, and the pose of the person in front of the camera — and still getting a good measurement. Robust anthropometry systems, therefore, need to employ calibration pipelines and normalization methods that take into account real-world variability and ensure that the measurements are comparable across sessions.

2.1.2 Computer Vision in Human Body Analysis

The human body has been revolutionized by computer vision, which has enabled applications from medical imaging to motion capture in entertainment. In particular, vision-based analysis of the human body enables systems to recognise the body's proportions, classify its posture and recognise movement patterns using simply the standard RGB image input. With the move away from depth cameras and dedicated hardware towards monocular vision-based approaches, these technology capabilities are now widely available and can be deployed without any requirement for infrastructure.

The perceptual backbone for fitness and muscle monitoring applications is computer vision. With large annotated datasets, the convolutional neural network can precisely segment body parts, locate each joint, and obtain silhouette information. Such capabilities are directly linked to pipelines for measuring the dimensions of the limb and monitoring limb variations over time. One of the most important challenges in this area concerns the transfer from 2D image perception to 3D anatomical reality: The human body is volumetric, but most readily available camera set-ups only capture a flat projection. To fill the gap, the challenge is to provide the appropriate geometric inferences and model-based estimates of meaningful 3D approximations that are extracted from 2D evidence by careful geometric inference.

2.1.3 Pose Estimation

Pose estimation is a mathematical process for identifying and finding the position of human body keypoints (usually joints like shoulders, elbows, wrists, hips, knees, and ankles) in an image or video frame. Most pose estimation systems use deep learning methods today, such as stacked hourglass networks, HRNet or transformer models that predict a heatmap or coordinate regression output for every keypoint. There are two general paradigms: top-down methods which first detect the person and then estimate the pose within each detected person box, and bottom-up methods which detect all of the keypoints globally, and then associate them to persons.

Pose estimation is the first step in a measurement pipeline in a hypertrophy tracking system. The geometric anchors, being segment lengths and circumference proxies, are accurately localized landmarks, especially those around the shoulder, elbow and wrist (upper body) or hip, knee and ankle (lower body). These estimation inaccuracies cascade through subsequent calculations and are critical to include confidence thresholds for keypoints as well as temporal consistency checks. Further, to be practically useful in real-world scenarios, pose estimation models should be able to be generalised to different body types, skin tones and clothing styles.

2.1.4 Geometric Modeling in Body Measurement

Geometric modeling is a mathematical model of physical objects that is simplified in terms of shapes and structure. Geometric modeling in the context of human body measurement is the process of approximating body parts and muscle groups by elementary geometric forms like lines, circles, cylinders and ellipses. Such a strategy allows transforming visual information into numerical values to offer an intermediate between the visual analysis and the physical one.

Linear distances are commonly employed in body measurement systems to depict lengths, e.g., the distance between two landmarks. The distances can be computed directly on the basis of coordinates of detected points. Nonetheless, linear measurements do not adequately represent the complexity of the human body, especially with regard to volume or cross-sectional area estimates. In a bid to solve this, several dimensions are fused together with the use of geometric formulas in an attempt to approximate more complicated structures.

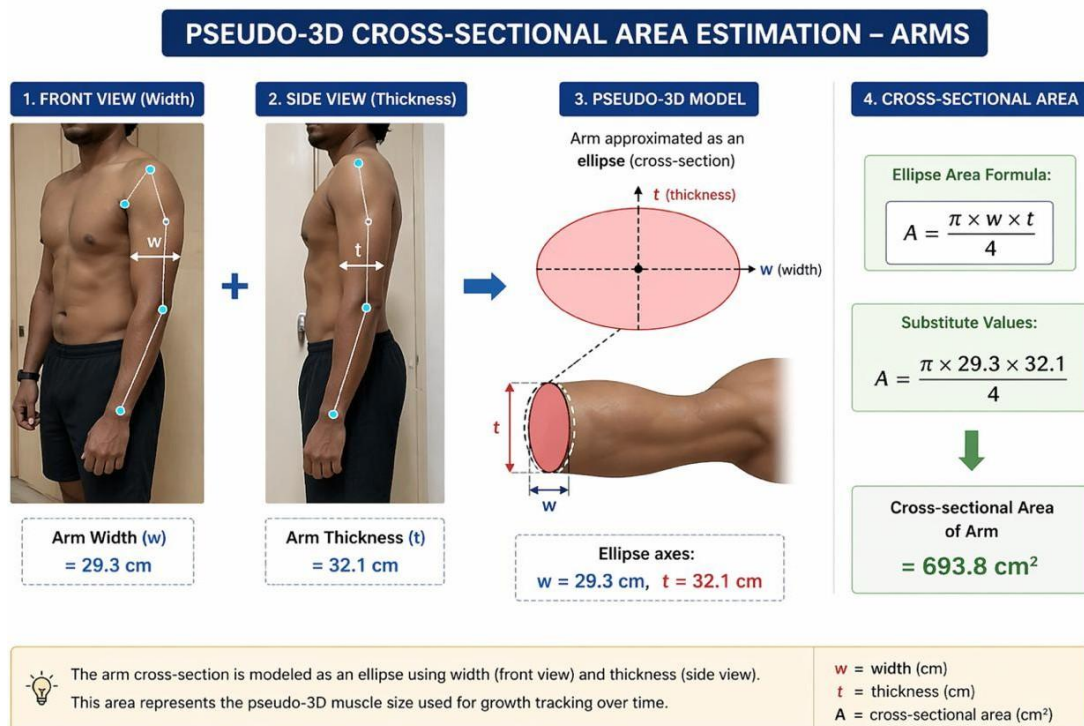


Figure 2. 1 Pseudo 3D Cross Girth Estimation

As an example, a part of the body, say the upper arm, can be modeled as a cylinder, with the length being the distance between landmarks of the shoulder and the elbow and the radius the circumference or width. The volume of the cylinder can be estimated using the normal geometric formulas. Likewise the cross section of a group of muscles can be estimated as an ellipse, where the major and minor axes indicate width and thickness respectively. The size of the ellipse gives an approximation of the size of the muscle which is also trackable with time.

Geometric models simplify the process of modeling complex anatomy, and it is possible to make calculations with the help of minimum data. Although these models do not take into account all the details of human body, they give a good approximation which is adequate in many of the applications. When it comes to measuring hypertrophy, it is not necessary to reach an ideal anatomical accuracy but rather to identify changes in muscle size in relation to time. Geometric modeling offers an effective and computationally easy method of doing this. Geometric modeling is also interpretable which is another strength of this modeling. The models have parameters that are of good physical meaning like lengths, radii and areas, which are simple to understand and analyze their results. This is especially critical in applications that are user-friendly, and clarity and transparency are crucial. Nevertheless, there is some assumptions and limitations in geometric modeling.

The models are accurate depending on the shapes that are used to depict the real anatomy. As an example, a model of a muscle as a perfect cylinder or ellipse might not represent changes in shape and curvature. Also, the input measurements are prone to errors, e.g., landmark positions, which can be spread over the model and influence the results. Geometric modeling is still a useful tool in body measurement and analysis in spite of these limitations. The proposed system is made more effective and reliable by using it with a correct landmark detection and multi-view data.

2.1.5 Limitations of Single-View Systems

The current body measurement systems that involve images used are based on one orientation; mostly the front or side of the target. Although this method makes the data acquisition and processing easy, it poses major limitations that influence the accuracy and completeness of the measurements. The main problem of the single-view systems is the absence of depth data, which is needed to comprehend the three-dimensional organization of the human body.

The entire spatial information is now projected onto a flat surface in a single 2D image, which loses one of the dimensions. This complicates the precise estimation of parameters like thickness, volume and actual spatial relationship among body parts. To illustrate, a front view can give the information on width but not on depth whereas a side view gives the information on depth and not on width. Consequently measurements based on a single view are incomplete in nature.

The second limitation is a lack of complete body structure. Human body is a complex three dimensional object that has different shapes and contours. This structure cannot be viewed in one perspective and thus it is partially or distortedly represented. Some of the features might be hidden or not seen at a certain angle, further lowering the accuracy of the analysis.

Perspective distortion also leads to a decrease in the accuracy of measurements in single-view systems. The distance of objects relative to the camera affects the apparent size of the object in the image hence they can be inaccurate without the correct consideration. These distortions are not easily corrected without some further reference points or calibration. Multi-view methods are needed to overcome these drawbacks. With images taken at various angles, one can get a more comprehensive view of the body. By using information in both front and side view, it is possible to estimate the width and depth and model the geometry more accurately. Multi-view systems also offer redundancy which can be utilized to authenticate measurements and enhance reliability.

To sum up, although single-view systems are easy to use and simple, they cannot be relied upon to measure depth and the entire body structure, which is why they cannot be used to measure hypertrophy accurately. Multi-view methods offer a more powerful solution, as they overcome these inadequacies and allow a more detailed analysis.

2.2 Reviewed Literature

The literature study in the field of image-based body analysis, pose estimation, and fitness tracking shows that there is a gradual development of the methods to measure and analyze the human body with the aim of automation. Although much progress has been achieved in the individual components, i.e. image processing, machine learning, and skeletal tracking, the combination of these components into a single system to track muscle hypertrophy has been underdeveloped. In this section, the existing research studies in various categories are critically analyzed and their strengths, limitations and the gaps that drive the proposed system are identified.

2.2.1 Body Measurement Systems based on images.

There is a large amount of literature on the problem of human body measurements in two-dimensional images based on computer vision and machine learning methods. The purpose of these systems is usually to approximate anthropometric measures like height, limb length and body circumferences without needing physical touch. The principle behind this is to identify body shapes or other important features and use geometric or statistical models to extract measurements based on pixel data. Most of these methods employ machine learning or deep learning models that are trained on annotated data to recognize relevant features on images. Convolutional neural networks (CNNs) are typically used in segmentation, where the body is divided out of the background, and keypoint detection, where particular points of the body are identified. Other systems also include regression models which can directly predict body measurements when fed with image inputs. These methods have proved it to be possible to extract body dimensions in images with a reasonable level of accuracy in controlled conditions.

The main benefit of image-based measurement systems is their non-invasive nature. The measurements can be accessed by the users without the need of physical contact and so the process is made more convenient and accessible. Moreover, these systems can be incorporated into mobile apps, making them widespread. The use of automation in measuring extraction also eliminates the use of human labor and errors.

Nevertheless, in as much as image-based systems have these benefits, there are several shortcomings. Among the most important issues is that most implementations do not have the capability of real-time processing. Although certain models are highly accurate, they may be very intensive in terms of computation, thus not applicable to real-time processes on regular devices. This reduces their application in interactive uses like live fitness tracking.

The other shortcoming is the relatively low level of accuracy of measurements in real life scenarios. Light differences, background interference, clothes and camera angles can greatly influence the functioning of these systems. In most instances, the models are educated using controlled datasets and might not be able to generalize to various environments. Consequently, the measurements derived are usually rough and might not be suitable in tracking accurately.

The most important limitation of the majority of image-based systems is the lack of longitudinal tracking functions. These systems are generally geared to measure one image or even a single session, and lack mechanisms of storing and analyzing data over time. This restricts their application in applications like hypertrophy monitoring where one needs to monitor variations at multiple time points.

2.2.2 Pose Estimation-Based Systems

Pose estimation-based systems is one of the major developments in human body analysis, which utilises the capability of the modern computer vision models to identify and track anatomical landmarks. These systems are concerned with the removal of skeletal structure of the human body based on the identification of keypoints including joints and limbs. The resulting skeletal representation is used to give a component structure to study posture, motion, and space relations among body parts.

Real-time feature is one of the prominent benefits of pose estimation-based systems. Frameworks like MediaPipe Pose and OpenPose are optimized to be run efficiently and thus can be used to process at high frame rates even on mobile devices. This renders them to be applicable in applications where immediate feedback is needed like in fitness coaching and activity monitors.

Another significant advantage is scalability. The model of pose estimation can be implemented on a broad spectrum of platforms, including smartphones and cloud-based solutions, enabling the model to provide its services to a significant number of users at the same time. Pre-trained models and standardized APIs make it even easier to integrate into applications.

Pose estimation systems are also characterized by lower computational complexity as opposed to entire 3D reconstruction techniques. They strike a balance between accuracy and efficiency by working on keypoints instead of dense 3D models. This renders them to be a viable option when limited computational resources are needed.

Although it has these benefits, there are some limitations in pose estimation-based systems. Their sensitivity to pose variation is one of the major challenges. Landmarks can be influenced by body orientation change, occlusion, and extreme poses which may lead to the inaccuracy of landmark detection. As an illustration, when a body part is partly obscured or has an abnormal arrangement, the model might not identify the respective keypoints correctly.

The other constraint is the reliance on proper landmark detection. All the latter measurements are based on these keypoints, so any mistake in the localisation of the keypoints may result in inaccurate measurements. This is of particular concern to applications that need accurate measurements, like hypertrophy monitoring. Also, pose estimation systems can only represent data in 2D unless they are augmented with 3D modelling methods. This limits their capacity to measure depth data which is critical to estimating the volumetric data like muscle size.

Although there are systems that seek to derive depth through machine learning, they usually make assumptions that might not always be accurate. To sum up, the systems of pose estimation offer an effective and strong tool of studying the structure of a human body. Their constraints in dealing with, however, demonstrate variation, depth estimation, necessitating complementary methods, including multi-view analysis, and geometric modeling, to provide more precise and complete measurement.

2.2.3 Fitness Tracking Applications

The rise of online fitness apps has changed the way people track their health and exercise. All these apps are usually a combination of sensor data, user-reported information, and wearable data to give information on different facets of fitness. Some of the common features are the calculation of calories, heart rate, the number of steps, and body composition.

The majority of the fitness apps are concentrated on the overall health indicators like the fat percentage of the body, body mass index (BMI), and the amount of activity daily. These measures can be helpful in evaluation of the general health and fitness but fail to give specific details regarding muscle development. As an example, BMI is an easy ratio of weight to height and fails to distinguish between muscle and fat mass. On the same note, body fat estimation methods give information on composition but not muscle size.

The other drawback of the current fitness apps is that they do not measure the muscle specifically. Although there are applications that enable the user to enter measurements like the arm or waist circumference, these are usually manual measurements, not part of automated analysis. Consequently, the individual tracking of muscle groups is restricted and usually not uniform. The current fitness applications also lack the use of geometric modeling extensively. Most of the systems do not seek to model body structure mathematically and this makes it hard to estimate volumetric changes. In the absence of such modeling, hypertrophy, or increase in muscle volume, is hard to accurately monitor, and not simply by linear dimensions.

Another area where the existing applications fail is in monitoring hypertrophy over a long period. Although numerous applications are available that give historical data on measurements like weight and activity level, they lack the ability to track muscle development over time in an organized way. This restricts their application to people who are interested in strength training and bodybuilding, where hypertrophy is a major objective. In short, although fitness tracking applications have come a long way in tracking overall health measurements, they are not as specific or analytical as needed in tracking muscle hypertrophy. This loophole underscores the necessity to have dedicated systems that can deliver detailed, automated, longitudinal analysis of muscle growth.

2.2.4 Multi-View Measurement Approaches

Multi-view measurement techniques are a major development in body analysis through images and some of the weaknesses of single-view systems have been overcome. Such techniques include taking photographs of the object in various positions, usually front, side, and even back. With the combination of the information presented on various perspectives, the more detailed representation of the body geometry can be gained.

The main benefit of multi-view systems is that they can acquire extra dimensions. Although one view can give only a small amount of information, it is the multiplicity of views that allows estimating the width, depth, and other spatial characteristics. It is essential to this especially in the accurate modeling of the three-dimensional structure of the human body. Multi-view systems have the potential to minimize ambiguity because they integrate data collected on various perspectives and enhance measurement accuracy. The other benefit is enhanced depth estimation. As in 2D images depth information is not directly accessible, multi-view methods exploit the geometric relationships between multiple views to extract depth information. This enables better estimation of volumetric characteristics, including muscle size. Multi-view systems in certain instances can estimate 3D reconstruction without having special hardware.

2.2.5 Research Gaps Identified

In the literature analysis, it is found that there are a number of serious gaps which impede the development of effective muscle hypertrophy tracking systems. These gaps emphasize the shortcomings of the existing approaches and give an effective direction to the suggested system. The lack of a specific system to track the longitudinal muscle hypertrophy is one of the most significant weaknesses. Although a large number of systems are capable of deriving measurements or approximate fitness indicators, only a small number are developed to monitor muscle development with time in a systematic and uniform fashion. This restricts the possibility of measuring training progress and making decisions based on data.

The other gap is that the current systems do not have pseudo-3D geometric modeling. The majority of methods are based on linear measurements or even basic metrics, without making an effort to estimate volumetric properties. As hypertrophy deals with the change of muscle volume, the lack of this modeling lessens the efficiency of these systems. The current research also pays very little attention to measurement validation. Most systems lack mechanisms to check the correctness of their output or consistency of results across sessions. This casts doubts on the dependability of the data and its appropriateness in analytical use.

Another field where the existing systems fail is the integration of the multi-view capture in easily manageable and affordable systems. Although the multi-view approaches have great benefits, they are normally adopted in complicated or resource-consuming settings. Solutions are required that would exploit multi-view data with the use of a conventional device like smartphones. Lastly, no accessible and scalable solutions exist to everyday users. There are numerous sophisticated systems but they are either costly, demand specialized apparatus, or too complex. This hinders their use and denies more people the opportunity to enjoy these technologies.

To sum up, the gaps in the identified research reveal the necessity of a system that will be both automatized, accurate, and scaled and allow users to use it. To fill these gaps, the proposed AI-based landmark-based hypertrophy monitoring system will combine pose estimation, multi-view analysis, geometric modeling, and longitudinal tracking into a single framework.

Ref	Title & Year	Methodology	Dataset	Limitations	Observations
[1]	2D Image-Based Digital Anthropometry Using Deep Learning (2024)	Deep learning-based landmark extraction from RGB images for body measurement estimation	RGB images of human subjects (single-session)	Focused on static body measurement; no longitudinal tracking	Demonstrates feasibility of 2D landmark-based anthropometry but lacks time-series modeling
[2]	Agreement of Anthropometric and Body Composition Measures Predicted from 2D Smartphone Images (2022)	Smartphone image-based body composition estimation compared with reference methods	Smartphone-captured 2D images with manual/DXA reference data	Focused on body fat & composition, not hypertrophy	Validates potential of 2D images for body metrics but not muscle growth tracking
[3]	Development and Validation of a Smartphone Application for Measuring Waist-to-Hip Ratio (2023)	Computer vision-based circumference estimation using smartphone images	Controlled smartphone image dataset	Limited to waist/hip metrics; no multi-region modeling	Shows smartphone CV can estimate circumferences reliably
[4]	Advances in Estimation of Body Fat Percentage Using AI 2D Photo Method (2025)	Deep learning regression model for fat % prediction from 2D photos	Large-scale 2D image dataset	Focused on fat estimation; not muscle-specific	Reinforces 2D viability but domain differs from hypertrophy
[5]	Prediction of Body Composition Using 2D Image-Derived 3D Meshes (2024)	2D images converted into approximate 3D body meshes for composition analysis	Multi-view RGB images with mesh reconstruction	Computationally heavy; not mobile-first	Highlights benefit of pseudo-3D modeling
[6]	Sex-Based Approach to Estimate Human Body Fat Percentage from 2D Images (2023)	Machine learning classification + regression from 2D images	Labeled dataset of male/female subjects	Focus on fat % classification; no longitudinal modeling	Confirms 2D can be used for predictive modeling
[7]	Smartphone-Based Human Pose Estimation for Anthropometric Measurement (2021–2023 studies trend)	Landmark-based pose estimation for body measurement	Controlled pose dataset	Static estimation; no time-based progression analysis	Landmark extraction is stable foundation for measurement systems

Table 2.2 Literature Survey Summary

2.3 Problem Formulation

The framework of this report is highly advanced to present a logical and detailed flow of ideas beginning with the fundamentals, to the specifics of the implementation and final findings. One of the chapters is a continuation of the other to ensure that the reader of the article is assured a clear picture on both theoretical and practical side of the proposed system. The framework shows the systematic approach to managing the problem-solving process, which is formed by context and analysis, design and implementation, as well as evaluation and future directions.

The problem formulation and literature review are the issues of Chapter 2. The chapter is a review of all the existing research on such topics as computer vision, pose estimation, digital anthropometry, and fitness tracking systems. It reviews the methodologies, strengths, and weaknesses of the current methods and what lacks in the current methodology is what the proposed system aims to address. The chapter uses the works of other scholars to build a strong theoretical framework and reasons why the proposed solution is needed. It also reduces the problem space by the definite formulation of the problems facing with the precise and automated monitoring of hypertrophy.

Chapter 3 is a formal and structured problem statement. This chapter will outline the specific issues that will be solved by the project, based on the findings of the literature review. It also determines the extent of the system, its objectives and measures of success. The problem statement plays a significant role between theory and practice, and it is a bridge, as a result of which the system design is associated with the objectives that have been set.

Chapter 4 expounds the system architecture and methodology which is the focus of the project. This chapter describes the overall layout of the system, the components, data flow, and computations. It outlines how input images are processed by pose estimation algorithms to obtain landmarks, how landmarks are used to calculate measurements, and how geometric modeling can be used to estimate muscle size. The chapter also discusses implementation details, such as preprocessing steps, model selection, calibration methods, and data storage mechanisms. It is possible to replicate this work with the help

of this chapter as it provides a thorough account of the methodology and learns more about the work of the system.

The results and discussion are presented in chapter 5 where the performance of the system is discussed with the assistance of the experimental data. This chapter includes quantitative measurements accuracy, traditional and existing digital methods comparison and visualization of the results of hypertrophy tracking. It also discusses the implications of the results, the strengths, and weaknesses of the system. The discussion provides sharp insights into the effectiveness of the proposed solution and proposes improvements.

The conclusion of the report is presented in chapter 6 and indicates scope in the future. In this chapter, the most significant contributions and findings of the project will be summarized and speculated on their significance in the context of the fitness technology and computer vision. It also covers potential future research and development paths that can be pursued such as enhancing the accuracy of the pseudo-3D modeling, real-time, or expanding the system to include other health and fitness indicators. The conclusion justifies the value of the project and emphasizes its potential to be improved.

Other sections such as appendices, references and acknowledgements can also be included in the report in addition to these important chapters. These sections provide more background information, details and citations to support the primary information. The overall organization ensures clarity, coherence and completeness that enable the reader to successfully navigate the report and have a complete picture of the proposed system.

This systematic way of reporting the hypertrophy tracking system based on AI will not only chronicle the way the system evolved but will also make a contribution to knowledge as a whole. It is both a source of technical knowledge and a source of further activity, demonstrating how advanced ways of calculating the answer to real-life problems in the sphere of fitness and health monitoring. Lack of consistency in measurement accuracy is another significant issue that has been noted in the existing methods. The performance of image-based systems can be greatly influenced by variability in the input conditions, including: lighting, camera angle, posture, and clothing.

To overcome these challenges, the proposed system presents a line of integrated solutions aimed at overcoming the limitations identified. The system has a multi-view landmark-based measurement framework at its core. The system records complementary data on width and depth by taking front and side pictures of the human body. This multi-view system helps to overcome the shortcomings of single-view systems and allows more thorough body geometry representation. The pose estimation is used to guarantee that anatomical landmarks are identified reliably, which forms a sound basis to extract measurements.

In addition to this, the introduction of a pseudo-3D geometric modeling approach is introduced. The system acknowledges the shortcomings of 2D imaging by using mathematical models to estimate the three-dimensional structures based on multi-view data. It builds simplified geometric models of muscle groups by combining measurements (e.g. width and thickness) which are typically modeled as shapes (e.g. ellipses or cylinders). This enables the system to approximate cross-sectional areas and deduce volumetric changes, which would give a more meaningful and accurate measure of hypertrophy than the traditional linear measures.

The validation-driven measurement framework is another important element of the proposed solution. The system has mechanisms to check the accuracy of measurements to guarantee reliability and consistency. This can involve cross-validation of multiple views, normalization methods to explain scale differences, and session to session consistency checks. The system minimizes the errors and increases the credibility of the data generated by integrating validation into the measurement pipeline.

The system also adds a longitudinal tracking system, which is necessary in tracking the growth of the muscles with time. The data acquired at various sessions are recorded in a systematic format, which allows analyzing the temporal trends. This data is presented in an intuitive way using visualization tools like graphs and progress charts, which enable users to easily interpret their progress. This feature will make the system no longer a mere measurement instrument but an active analytical platform.

Together, these elements are designed to deliver some of the important advantages. To begin with, multi-view analysis and geometric modeling are integrated, which results in better measurement accuracy. The system gives a more realistic representation of muscle size by representing more dimensions and estimating the volumetric properties. Second, the measurement extraction and validation is automatized, which ensures consistency between sessions, minimizes variability, and improves reliability. Third, the longitudinal tracking mechanism makes it possible to analyze the progress in a meaningful way, letting users track their progress and make informed decisions regarding their training.

Finally, the problem formulation indicates the serious gaps in the current systems and defines the necessity of a multi-faceted solution that incorporates several highly developed techniques. The suggested system fills these gaps by using a multi-view analysis, pose estimation, geometric modeling, validation, and longitudinal tracking. This combination methodology not only addresses the shortcomings of the existing techniques, but also offers a solid structure to precise, reliable, and scalable muscle hypertrophy monitoring.

3 Problem Statement

Preamble to the Chapter:

This chapter is the official definition of the problem that the proposed AI-based landmark-driven muscle hypertrophy tracking system is going to address and the basis upon which it is going to be designed and implemented. It converts the information obtained during the literature review into a clear and organized problem statement, outlining the shortcomings of the current solutions and the need to develop a more sophisticated solution. Moreover, this chapter defines the main purpose of the project and identifies the objectives needed to fulfill the purpose. It also defines the extent of the system, where the limits of the development and implementation of the solution lie.

This chapter is important because it is a transition between theory and practice. It is also used to ensure that the system design and methodology that follows are aligned with clearly defined requirements by clearly defining the problem and related objectives. Such an organized methodology helps in maintaining clarity, consistency and focus during the development process such that the project can respond to real-life challenges in a systematic and efficient way.

3.1 Problem Statement

The application of artificial intelligence and computer vision to the field of fitness and health monitoring has resulted in the creation of a broad assortment of digital tools in the recent years that are intended to enhance the way people evaluate and monitor their physical state. Such technologies have facilitated automation, remote access and data-driven insights, greatly improving the user experience in aspects like tracking activities, monitoring calories, and overall health evaluation. With all these developments, there is one area of fitness that is lacking, which is proper monitoring of muscle hypertrophy. One of the main objectives of those who practice strength training and bodybuilding is muscle hypertrophy, or the growth of muscle size as a result of resistance training. Nevertheless, lack of reliable and automated systems of measurement remains a significant challenge in measurement and monitoring of this type of physical development.

Conventional techniques of monitoring muscle development are still popular, mainly because they are simple and inexpensive. Manual measurement where measuring tapes are used to measure the circumference of certain body parts like arms, chest, waist and thighs is one of the most common methods. Although this is a simple way of giving a quantitative value, it is very reliant on the accuracy and consistency of the person conducting the measurement. The inconsistency of the results can be caused by variations in the tape placement, tension, posture, and measurement technique and it is not easy to be sure that the changes will be tracked over time. Moreover, manual measurements are usually taken at non-regular time points and are not standardized, which further diminishes their usefulness to analyze over time.

Visual comparison is another method that is frequently used, and people estimate their progress by comparing photos taken at various times. Such method is subjective by nature and depends greatly on human perception, which can be affected by numerous external factors. Lighting, camera angle, background and body posture differences can make a big difference in muscle definition and size. Consequently, the use of visual comparison can be highly misleading and cannot be as precise as it is needed to track hypertrophy. Moreover, subjective evaluation does not have any quantifiable data and, therefore, cannot be analyzed systematically or incorporated into electronic systems.

Besides these conventional approaches, available digital fitness systems have tried to cover areas of body measurement and health monitoring. The most popular metrics on these systems are Body Mass Index (BMI), body fat percentage, and static body measurements. Although these indicators are good in evaluating overall health and body composition, they do not directly determine muscle size or growth. BMI, one example, is a simple weight to height ratio, and fails to distinguish between muscle and fat mass. On the same note, body fat estimation gives a clue of composition, but fails to provide a detailed data on individual muscle groups or their development.

In addition, the majority of the current digital systems are confined to the concept of static measurements, which record data at one time, but lack mechanisms of continuous or longitudinal tracking. This weakness does not allow users to study trends and the way their body changes with time. The time-tracking capability is crucial in the context of

hypertrophy, where the change is slow and cumulative. In the absence of longitudinal data, users are not able to assess the effectiveness of their training programs or make informed modifications. The other severe shortcoming of existing systems is the inability to measure in multi-views. Most image-based methods are based on one image, usually a frontal picture, to determine body sizes. Nonetheless, human body is naturally in three-dimension and one two-dimensional picture cannot be used to obtain depth data. This causes the lack of full representation of body structure especially in terms of thickness and volume. As a result, the measurements based on single-view systems are not accurate and cannot be used to estimate the size of the muscle, which is not only dependent on the linear size but also on the volume characteristics.

Besides lack of depth estimation, there are also no effective mechanisms of validating measurements in the existing systems. The extracted measurements cannot be checked to be accurate or consistent between different sessions without validation. Unreliable results can be caused by factors like changes in image capture conditions, landmark detection errors and scaling errors. Lack of validation structures diminishes user trust and constrains the real-world usability of such systems.

Moreover, the majority of existing methods lack the use of geometric modeling methods to model muscle structure. The growth of muscles is a three dimensional phenomenon with volume and cross-sectional area changes. But it is hard to convert measurements based on images into significant representations of muscle size without proper modeling. Existing systems cannot give accurate and interpretable insights on hypertrophy due to the absence of geometric modeling.

Another major shortcoming is associated with the use of single-view image-based systems, which do not reflect the entire spatial nature of the human body. As these systems are based on 2D projections, they cannot consider the depth, thus providing distorted or incomplete measurements. This is particularly important when one is trying to estimate the size of the muscle since it involves both the thickness and the width. Single-view systems are not able to give a complete analysis of muscle structure without the inclusion of multiple perspectives or modeling techniques.

3.2 Problem Definition

Based on the identified limitations, the core problem can be formally defined as the lack of an automated, accurate, and scalable system capable of extracting body measurements, estimating muscle size, and tracking hypertrophy over time using accessible technologies. In order to solve this issue, it is evident that a system that can carry out the following functions is required:

First, the system should be able to extract body measurements by applying computer vision methods. This is done by examining photographs of the human body to identify important anatomical features and calculate distances and sizes using these features. It should be completely automated, and the process requires minimal human intervention and consistency in measurements.

Second, the system should be able to estimate muscle size correctly. As hypertrophy of muscles is a matter of volume, the system must include techniques of estimating three dimensional structures based on two dimensional data. This necessitates application of geometric modeling methods that can model muscle groups in a simplified but significant manner.

Third, the system should facilitate monitoring of muscle growth with time. This involves the capability to save measurements across different sessions, arrange them in a systematic way, and examine trends. Longitudinal tracking plays a crucial role in the process of comprehending the progress and determining the effectiveness of training programs.

Fourth, the system should be able to give consistent and reliable results. This includes the use of validation mechanisms to make sure that measurements are correct and comparable between sessions. Stability is essential to gain user trust and conduct meaningful analysis. To conclude, the problem statement shows that a holistic solution is required that will combine computer vision, geometric modeling, and data analysis into a single framework. This type of system must address the shortcomings of current methods and offer a powerful platform on which muscle hypertrophy can be accurately and automatically tracked.

3.3 Title, Aim and Objectives

The title, aim, and objectives formulation provide a clear and well-organized direction of the proposed system and the necessary alignment of the development process with the identified problem of inaccurate and non-automated muscle hypertrophy tracking. This section translates the constraints that have been witnessed in the current systems to a clear solution framework. It gives an understanding of what the system is going to accomplish and how every part of the system helps in resolving the issue. By specifying these elements in a systematic and elaborate way, the project guarantees uniformity in design, implementation and evaluation.

Title

The project is called AI-Based Multi-View Landmark-Driven Muscle Hypertrophy Tracking System with Longitudinal Growth Modeling.

The title is indicative of the combination of computer vision and artificial intelligence in studying the human body. The AI-Based term means that the system is based on intelligent algorithms to process and interpret visual data. Multi-View emphasizes the application of multiple views, that is, front and side views, to address the shortcomings of single-view analysis of images. The term Landmark-Driven underlines the fact that the system relies on anatomical keypoints identified by the estimation of poses as the basis of the measurement extraction.

The title of the project, which is the Muscle Hypertrophy Tracking System, makes it clear what the project aims to accomplish but it does not aim at measuring and monitoring fitness but muscle growth specifically. Lastly, the system is characterized by the term Longitudinal Growth Modeling, which suggests that the system is not only meant to measure muscle size but also analyze the changes, allowing users to monitor the progress and discern the growth patterns. The title, thus, reflects both the technical approach and the field of application of the system.

Aim

This project is intended to design and develop a computer vision-based system capable of automatically quantifying body size, estimating muscle size through geometric modeling methods, and monitoring muscle size change over a period of time through multi-view landmark detection. This objective is dedicated to filling in the main gaps of the current systems with introducing automation, accuracy, and scalability to hypertrophy tracking. The conventional ways are very manual and depend on subjective interpretation, making them inconsistent and not precise. Through computer vision, the proposed system will not require manual measuring and will be able to extract the body dimensions automatically using a computer vision.

The estimation of muscle size through geometric modeling is an important fact about the aim. As muscle hypertrophy is a change in volume, it cannot be effectively measured by linear measurements. This is taken care of in the system by estimating the mathematical models of the muscle structures and this enables the size of the muscle to be approximated more realistically.

Multi-view analysis, which is critical in measuring both width and thickness of the body, is also included in the goal. Using a combination of information on the front and side views, the system can estimate three-dimensional features based on two-dimensional images. The objective further focuses on longitudinal tracking, which enables the system to track the changes through time and give significant information about muscle growth. This changes the tracking process to a dynamic analytical system as opposed to a static measurement approach.

Objectives

In order to attain the aforementioned goal, the project will be broken down into a number of objectives with each objective concentrating on a particular element of the system.

The former aims at estimating anatomical features by pose estimation methods. This includes the deployment of a robust pose estimation system that is able to identify important body parts including shoulders, elbows, hips, and knees. MediaPipe Pose is used in the system because of its high accuracy and real-time performance. These marks are used to extract measurement and are used to guarantee consistency between sessions.

The second one is to adopt a multi-view measurement system. The system records the front and side of the human body to get a complementary record on the body dimensions. Width related measurements are taken using the front view and thickness is obtained with the help of the side view. The combination of these two views results in the system transcending the limitations of single-view systems and provides a more comprehensive representation of body structure.

The third goal is to come up with a pseudo-3D geometric model to estimate muscle. As the 2D images cannot be directly used to measure the size of the muscle, the system employs the geometric modeling methods to estimate the muscle size. Specifically, muscle cross-sections are modeled in the form of an ellipse, with the width and thickness measurements determining the axes of the ellipse. This enables the system to approximate cross-sectional area and deduce volumetric changes with hypertrophy.

The fourth goal is to develop a measurement reliability system. This includes adopting validation systems that will confirm that measurements are accurate and consistent. Pose validation is utilized in checking whether the user is in a standard posture at the time of capturing the image and consistency checks are done across sessions to identify any differences. These will assist in minimizing the amount of errors and enhance the dependability of the system.

The fifth goal is to conduct longitudinal growth monitoring. The system captures data on measurements over a period of time in a systematic format and allows analysis of variations over a number of sessions. The system will be able to detect trends and patterns in muscle growth by comparing past and present measurements. This data is presented in an intuitive way with the help of visualization tools like graphs and charts.

The sixth goal is to introduce elementary growth modeling. This includes the use of regression techniques to examine the pattern of measurements and approximate patterns of growth. The system can offer predictive information to users by modeling muscle size evolution with time, which can help them assess the effectiveness of their training.

The seventh goal is to verify accuracy of systems. This is done by contrasting results of the measurements of the system and that of the manual methods like measuring tapes. This validation process assists in determining the correctness of the system and makes the results of the system reliable and consistent.

All in all, the title, aim, and objectives are all a well-organized plan to create an automated, precise, and scalable muscle hypertrophy tracking system.

3.4 Scope of the Project

The scope of the project defines the boundaries within which the proposed system is designed, developed, and evaluated. It clarifies what functionalities are included, what limitations are intentionally accepted, and where the system can be practically applied. Establishing a well-defined scope is essential to ensure that the project remains focused, feasible, and aligned with its objectives. It also helps in setting realistic expectations regarding the capabilities of the system and avoids overextending into areas that require specialized resources or fundamentally different methodologies. The proposed AI-based muscle hypertrophy tracking system is primarily centered on leveraging computer vision and geometric modeling techniques to provide an accessible, automated, and scalable solution for tracking muscle growth. The scope is therefore defined in three parts: in-scope functionalities, out-of-scope limitations, and application areas.

In Scope

The system focuses on upper body and lower body measurements as the primary regions of interest for hypertrophy tracking. These include major muscle groups such as the arms, chest, shoulders, back, thighs, and calves. By targeting these regions, the system is able to provide meaningful insights into muscle growth while maintaining computational simplicity. The selection of these body segments is also aligned with common fitness practices, where hypertrophy is typically monitored in these areas.

A key component within the scope is landmark-based measurement extraction. The system relies on pose estimation techniques to detect anatomical landmarks such as joints and reference points on the body. These landmarks serve as the basis for calculating distances and dimensions, ensuring that measurements are consistent and repeatable across sessions. This approach eliminates the need for manual measurement and reduces variability caused by user-dependent factors.

The system also incorporates multi-view capture using a single webcam. Instead of requiring multiple cameras or complex setups, the system captures front and side views sequentially using a standard camera. This design choice enhances accessibility and scalability, allowing users to perform measurements using commonly available devices such as laptops or smartphones. By combining information from multiple views, the system is able to approximate both width and thickness, addressing the limitations of single-view analysis.

Another important aspect within the scope is pseudo-3D muscle estimation. Since the system operates on 2D images, it does not directly capture three-dimensional data. Instead, it uses geometric modeling techniques to approximate muscle size. For example, muscle cross-sections are represented using simplified shapes such as ellipses, with dimensions derived from multi-view measurements. This approach enables the estimation of cross-sectional area and provides a more meaningful representation of muscle growth compared to linear measurements alone.

The system also includes longitudinal tracking and visualization. Measurements obtained from different sessions are stored in a structured format, allowing for the analysis of

changes over time. Visualization tools such as graphs and progress charts are used to present this data, enabling users to monitor their growth and identify trends. This feature transforms the system from a static measurement tool into a dynamic analytical platform, supporting long-term fitness tracking.

Out of Scope

While the system provides several advanced capabilities, certain functionalities are intentionally excluded to maintain feasibility and focus. One of the primary exclusions is medical-grade body composition analysis. The system is not designed to replace clinical tools used for measuring parameters such as bone density, muscle mass distribution, or internal body composition. Such analyses require specialized equipment and standardized procedures that are beyond the scope of this project.

The system also does not provide exact volumetric muscle measurement. Although it estimates muscle size using geometric modeling, these estimates are approximations rather than precise measurements. The complexity of human anatomy and the limitations of 2D imaging make it impractical to achieve exact volume calculations without advanced 3D scanning technologies. Therefore, the system focuses on relative changes and trends rather than absolute precision.

Advanced biomechanical modeling is another area that is excluded from the scope. The system does not analyze muscle mechanics, force generation, or movement dynamics. While pose estimation provides information about body structure, it is not extended to detailed biomechanical simulations or performance analysis. Incorporating such features would require additional data sources and computational models that are beyond the current objectives.

The system also does not include multi-camera 3D reconstruction. While multi-view analysis is used, it is implemented using a single camera capturing different perspectives rather than a synchronized multi-camera setup. Full 3D reconstruction typically requires multiple cameras, depth sensors, or specialized hardware, which would increase complexity and reduce accessibility. The proposed system prioritizes simplicity and usability by relying on widely available devices.

Application Scope

The proposed system has a wide range of practical applications, particularly in domains related to fitness and health monitoring. One of the primary applications is fitness tracking, where users can monitor their muscle growth and overall physical development. By providing objective and consistent measurements, the system enables users to evaluate their progress and stay motivated.

Another important application is strength training analysis. Individuals engaged in resistance training can use the system to assess the effectiveness of their workout routines. By tracking changes in specific muscle groups, users can identify which areas are improving and adjust their training programs accordingly. This data-driven approach enhances the efficiency and effectiveness of training.

The system can also be used for personal health monitoring. While it is not a medical device, it provides valuable insights into physical changes that may be relevant for general health and wellness. For example, users can track changes in body composition and identify trends that may indicate improvements or concerns.

In addition, the system has potential applications in research, particularly in the field of digital anthropometry. Researchers can use the system to collect and analyze data on body measurements and growth patterns. The automated and scalable nature of the system makes it suitable for large-scale studies, where manual measurement methods would be impractical.

Overall, the scope of the project is carefully defined to balance functionality, accuracy, and accessibility. By focusing on key features such as landmark-based measurement, multi-view analysis, geometric modeling, and longitudinal tracking, the system provides a practical and effective solution for muscle hypertrophy tracking while maintaining clear boundaries on its capabilities.

3.5 Methodology Mapping to Objectives

The methodology mapping provides a structured representation of how each objective of the proposed system is implemented using specific techniques and tools. It establishes a direct relationship between the conceptual objectives and their practical execution, ensuring clarity in system design and development. Each objective is supported by an appropriate methodology and corresponding tools or resources, enabling a systematic and organized approach to implementation.

Module	Objective Addressed (What)	Method / Architecture Used (How)	Justification (Why)
Image Capture Module	Capture standardized physique images	Mobile-based front and side pose capture interface	Multi-view capture enables pseudo-3D geometric approximation and reduces depth-related distortion
Landmark Extraction Engine	Identify anatomical key points	Pose estimation model (e.g., landmark-based skeletal mapping)	Provides consistent coordinate mapping for measurement computation
Pose Quality Verification	Ensure posture consistency	Shoulder alignment check, hip alignment check, tilt angle threshold validation	Prevents geometric distortion caused by improper body orientation
Scale Normalization Module	Convert pixel values to real-world units	Scaling factor = $\text{real_height} / \text{detected_pixel_height}$	Eliminates camera distance variation effect
Multi-Frame Averaging	Reduce landmark detection noise	Capture 3–5 frames and compute mean measurement	Improves stability and reduces random detection variance
Pseudo-3D Geometric Modeling	Approximate muscle thickness and cross-sectional change	Combine width (front view) and thickness (side view); apply ellipse-based area estimation	Provides lightweight alternative to full 3D reconstruction
Measurement Engine	Compute muscle-related metrics	Distance computation between normalized landmarks	Enables region-specific hypertrophy tracking
Muscle Hypertrophy Index (MHI)	Provide composite hypertrophy score	Weighted aggregation of normalized region growth metrics	Offers holistic growth representation across multiple regions
Longitudinal Storage Module	Maintain time-series progression	Database storage with timestamped records	Enables longitudinal hypertrophy modeling
Growth Modeling & Prediction	Estimate growth rate and future trend	Regression-based time-series modeling	Converts tracking into predictive analytics
Statistical Validation Layer	Evaluate measurement reliability	Manual vs AI comparison; MAE, RMSE, correlation analysis	Provides scientific accuracy validation and defensible error bounds
Mobile Deployment Architecture	Enable real-world product scalability	Mobile frontend + backend processing pipeline	Ensures system applicability beyond research environment

Table 3. 5 Methodology

4 Problem Solving

The project execution plan was significantly refined based on the feedback received during the Interim Project Presentation, marking a critical transition from a conceptual prototype into a scalable, implementable, and intelligent fitness analytics system. The primary objective established after the interim review was to migrate from a prototype-level proof of concept to a full-stack, production-oriented architecture capable of supporting real-time AI-based body measurement and fitness analytics.

As part of this transition, the system architecture was redesigned around a modular full-stack framework. A React-based frontend application was developed to handle all user-facing interactions, while FastAPI was adopted as the backend framework responsible for API handling and computational services. Supabase was integrated as the cloud platform to manage authentication and persistent data storage. This architectural migration was carried out to replace earlier mock implementations with a deployment-ready system capable of scaling effectively under real-world usage conditions.

A major enhancement introduced following the interim review was the integration of a Real-Time Computer Vision Measurement Module. This component was designed to leverage live webcam input to perform continuous pose landmark detection using MediaPipe Pose, enabling the system to move away from static or manually provided image inputs toward a fully automated and intelligent measurement workflow planned was the ability to view live computer vision data with webcam input.

The suggested implementation involved the following:

- Real-time pose landmark detection with MediaPipe pose.
- System for automated body posture validation.
- It allows to use life-scale overlay for user instructions
- As for precise pose estimation, it is essential to keep obtaining the landmarks continuously.

Multi-View Measurement Workflow

A multi-view capture workflow was designed to increase the reliability of capture and estimate dimensions accurately.

The multi-view measurement workflow consisted of several coordinated stages. It began with an advanced self-adjusting front body position detection mechanism that automatically validated the user's posture before proceeding. Once stability was confirmed through pose validation, the system triggered a front-view capture. The user was then guided to transition to a side-facing position, after which a side-view capture was performed. The landmark data collected from both views were subsequently fused together to produce the final body dimension estimation. This multi-view approach was deliberately chosen over single-frame estimation to yield more robust and dimensionally accurate results.

Following the refinement of the project objective after the interim review, the measurement scope was narrowed specifically toward muscle hypertrophy tracking rather than generic overall body fitness metrics. The focus was directed at a defined set of measurable parameters: shoulder width estimation, arm girth estimation, and thigh girth estimation. Each measurement was accompanied by a confidence score to establish the reliability of the recorded values. This deliberately limited scope was chosen to align the system precisely with the domain of strength training and physique progression analysis, ensuring that the measurements produced were meaningful and actionable for users engaged in structured muscle development programs.

Application Security and Management

A robust and secure authentication workflow was planned to manage user access across the platform. The implementation covered user registration, a secure login system, JWT and session-based authentication mechanisms, and protected API access restricted exclusively to authorized users. Cloud-based authentication integration was planned through Supabase services to ensure a reliable and scalable identity management foundation for the system.

Database Integration and Persistent Data Storage

A persistent cloud database architecture was designed to support long-term storage and management of user data. The storage model was structured to preserve a comprehensive set of records including user profile information, measurement session records, historical body measurement data, growth trend analysis records, diet recommendation data, and user activity logs. This robust data retention framework was intended to facilitate longitudinal progress tracking and enable the delivery of personalized analytics tailored to each user's fitness journey.7. Dashboard and Analytics Module

Data processing would feed into a personal user dashboard to display processing progress and insights.

Some of the things that were suggested for the Dashboard would be:

- Latest measurement summary
- Historical progress visualization
- Growth trend analytics
- Prediction insights
- Profile summary
- Session history

This was supposed to enhance the usability and user engagement of this module.

Predictive AI Module

An AI-assisted prediction component was planned to provide forward-looking hypertrophy insights for users. The system was designed to perform historical trend analysis, generate muscle growth predictions, produce progress forecasts, and deliver confidence-assisted prediction outputs. This feature was intended to meaningfully enhance the overall intelligence of the platform by enabling users to make informed decisions based on projected physiological trends rather than relying solely on historical data.

In addition to the physical measurement capabilities, a Personalized Diet Recommendation Module was designed to support holistic fitness progression. The proposed functionality centered on analyzing existing measurement data to identify nutritional requirements aligned with the user's hypertrophy objectives. From this analysis, the module was intended to generate macronutrient intake guidelines covering protein, carbohydrate, and fat targets. Meal plan suggestions tailored to the user's current body composition phase and training intensity were also included in the planned scope. Furthermore, the module incorporated a dynamic recalibration mechanism to continuously update dietary recommendations as new measurement sessions recorded changes in the user's physical progress over time. This module was conceived as a bridge between physical tracking and nutritional guidance, transforming the platform from a standalone measurement tool into a fully integrated and connected fitness management experience.

Creation of Responsive User Interfaces and UI/UX design.

The complete redesign of the frontend interface was planned to guarantee the professional usability and accessibility on all devices.

A comprehensive frontend interface redesign was planned to ensure professional usability and accessibility across devices. The UI/UX enhancements adopted a modular component-based design using React, with clearly separated views for measurement, dashboard, and profile management. The layout was optimized to be responsive across both desktop and mobile screen dimensions. Real-time visual feedback was incorporated during pose detection and measurement capture to keep users informed throughout the process. An intuitive navigation flow was designed to guide first-time users through the measurement workflow without requiring any prior technical knowledge. Color schemes and typography were carefully harmonized with established fitness application design principles. Overall, the interface redesign was intended to minimize user friction and ensure the system could be operated independently by non-technical users.

A structured integration and testing phase was designed to validate the complete system pipeline from frontend interaction through to backend computation and database

storage. The planned testing activities covered unit testing of individual backend API endpoints using FastAPI's test client, integration testing of the computer vision pipeline to verify landmark detection accuracy across varied lighting conditions and postures, and end-to-end workflow testing spanning user registration, measurement session execution, data storage, and dashboard retrieval. Boundary condition testing was also planned to evaluate system behavior in edge cases such as partial body visibility and low-confidence pose detections. Performance testing under concurrent user load was included to assess API response latency and real-time measurement throughput. This testing phase was designed to confirm functional correctness, system stability, and measurement reliability prior to final evaluation.

A cloud-based deployment plan was formulated to make the system accessible beyond the local development environment. The strategy involved deploying the FastAPI backend service on a cloud hosting platform with secure environment variable management for API key handling. The frontend was published via a static hosting service integrated with continuous deployment support. Supabase project credentials were configured for production-level database and authentication access, and domain routing was established to ensure seamless communication between the frontend and backend services. This deployment plan was designed to demonstrate the system as a viable, production-ready application rather than a restricted academic prototype.

In parallel with system development, comprehensive documentation and codebase organization were planned to support both academic submission and potential future development. The documentation activities included inline code comments and modular function-level documentation across the frontend, backend, and computer vision pipeline components. API endpoint documentation was generated using FastAPI's built-in OpenAPI specification. System architecture diagrams were prepared to illustrate data flow between the frontend, backend, and cloud services. A user guide was drafted to outline the measurement workflow for end users. Finally, the complete project report was compiled integrating all chapters, results, figures, and references in full compliance with university submission guidelines.

4.1 System Architecture

The proposed AI-Based Multi-View Landmark-Driven Muscle Hypertrophy Tracking System is built on a systematic and modular pipeline model which guarantees a high level of accuracy in measuring the extraction of measurements, repeatability of measurements across different sessions and future upgrading of the system. The architecture combines both computer vision and geometric modeling with data analytics into one comprehensive system that converts raw visual data into valuable hypertrophy insights.

A high degree of automation and reliability is built into the system. This architecture, unlike the traditional system where the measurements or subjective interpretation are done manually, makes sure that every step of the processing is processed in a computational manner with minimum human intervention. Each component can operate on its own and can contribute to the overall pipeline as it is modular, which allows flexibility in upgrading or altering individual modules without impacting the entire system. This design philosophy is more so when it comes to research-based and dynamic systems wherein the improvements are ongoing.

On a high level, the system is based on a pipeline model. Each phase carries out a clear-cut operation, which transforms the input information into systematic products. The pipeline will reduce the number of mistakes and trace the transformations and consistency of the results in the various sessions. Moreover, the architecture is extensible, and more sophisticated methods like deep learning-based modeling, cloud data storage, or real-time analytics can be added in the future.

Overall System Flow

The system end-to-end workflow can be modeled as:

User pose estimation landmark extraction measurement module scaling module multi-view fusion pseudo-3d modeling data storage visualization.

The transformation of the input data at each stage is vividly presented in this workflow. Raw video frames are first captured and afterwards processed to obtain landmarks. Measures are calculated based on these landmarks and are normalized, combined across

views and transformed into useful geometric representations. Lastly, the processed data will be stored and visualized to be interpreted by the user. The architecture diagram will be included in the report as shown below: (Insert architecture diagram here).

The architecture consists of six large layers:

1. Input Acquisition Layer
2. Pose Estimation Layer
3. Measurement & Scaling Layer
4. Multi-View Fusion Layer
5. Pseudo-3D Modeling Layer
6. Storage & Visualization Layer

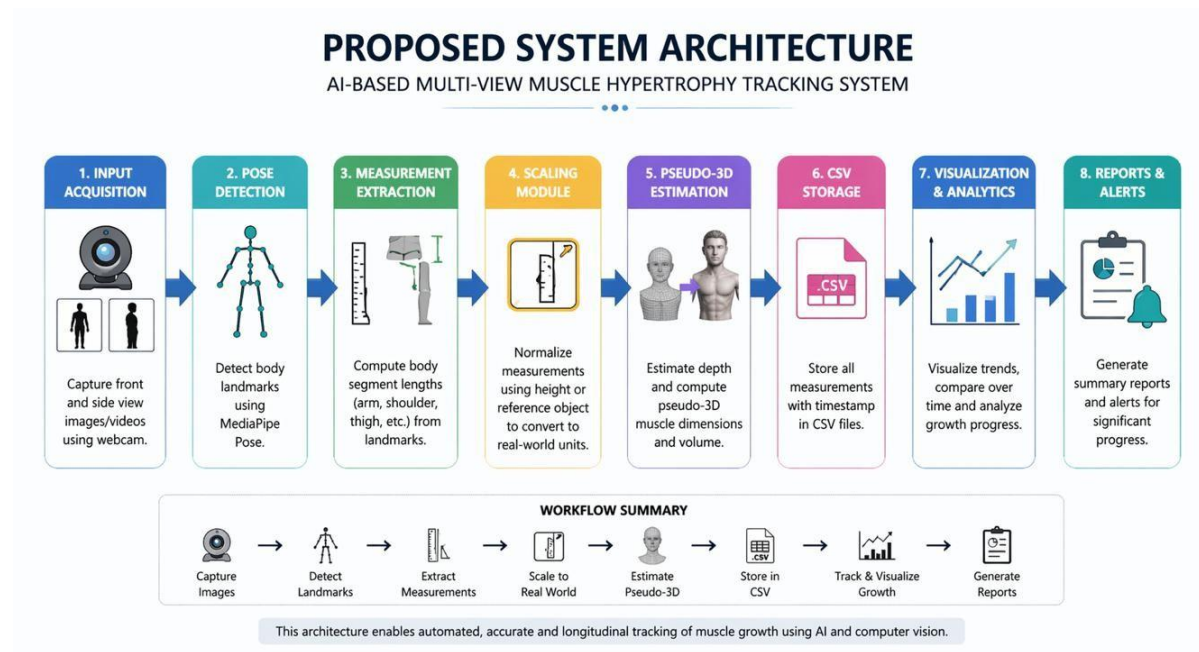


Figure 4. 1 System Architecture

All these layers are important in giving the system an efficient and accurate functioning.

1. Input Acquisition Layer

The input acquisition layer is used to gather visual data of the user. The system is based on a standard webcam or camera device and is therefore very accessible and does not require expensive or specialized hardware. The design makes sure that the system can be easily implemented in real-life settings like at home, gym, or a research lab. There are two kinds of captures that the user must execute:

- Front View (F): Records the frontal view of the body, mostly applied to measure width-based parameters like shoulder width, arm width and torso spread.
- Side View (S): The lateral view, which is used to determine the thickness-related parameters, e.g., chest depth, limb thickness.

This two-view capture system is crucial in estimating the three-dimensional body features with two-dimensional inputs. The system fills the complementary perspectives with depth information that is missing in single-view systems. To provide consistency and reliability, the system has a number of constraints during capture. These involve using a fixed position (sitting upright with the arms slightly apart), minimal camera movement, and sufficient and even lighting conditions. The system can also contain on-screen instructions or guides to assist users to get themselves into the right position in the camera frame. These steps minimize variations and enhance the quality of the next processing steps.

2. Pose Estimation Layer

The pose estimation layer takes the captured frames, and it identifies the anatomical landmarks. This is done through the MediaPipe Pose framework that is generally known to be efficient, real time and highly accurate in the detection of human body keypoints. The system identifies the important anatomical landmarks that are applicable in measurements which include: Left Shoulder and Right Shoulder. Left Elbow and Right Elbow. Left Hip, Right Hip. Knee and Knee Left. Every landmark is defined as a point in three-dimensional space: (x, y, z)

where:

- (x, y) are normalized coordinates of the frame of the image. Where (z) is relative depth information. Even though the system mainly works with 2D data, the (z)-coordinate gives extra contextual data that can be used to refine measurements. The pose estimation is done in real time where the system can give instant feedback and proper user positioning before measurements are taken. This layer is essential since the rest of the computations will rely on the precision of identified landmarks. Landmark detection errors may cause wrong measurements and thereby disrupt the system reliability. As such, this layer is meant to be resistant to changes in lighting, background, and user appearance.

3. Measurement & Scaling Layer Once the landmarks have been identified, the system calculates distances between certain pairs of landmarks by using the Euclidean distance. These are bodily physical dimensions. Examples of measurements are:

- Arm Length: Shoulder-elbow distance. Shoulder Width: Space between left and right shoulders.

- Thigh Length: Hip to knee distance. These measurements are first expressed in units of pixels. The pixel values are however dependent on the camera resolution and the distance and thus cannot be compared in the real world. A scaling mechanism is used to transform these measurements into units of the real world. The system takes the height of the user as a reference to calculate a conversion factor:

$$cm_per_pixel = \frac{0.93 \times \text{user height}}{\text{pixel height}}$$

This scaling makes sure that measurements are normalized and do not depend on camera settings. The adjustment factor of 0.93 is based on the normal human body proportions. Using this conversion, the system will be able to convert all measurements to the same units (centimeters) and then compare accurately across the various sessions and users.

4. Multi-View Fusion Layer

Multi-view fusion layer is used to provide a more complete body size representation by combining the measurements of front and side view. In this layer: Front view gives width measurements. Thickness measurements are given by side view. Such measurements are synchronized and combined to estimate the three dimensional body structure. As an illustration, the thickness of a muscle group measured on the front view is added to the width of the same group, and the thickness of the same group measured on the side view is added to the width of the group to create a more precise image. This is the solution to the key weakness of single-view systems, as they are unable to capture depth information. The system enhances the accuracy of measurements by combining several views and allows analysis of muscle size to be done more meaningfully.

5. Pseudo-3D Modeling Layer

The pseudo-3D modelling layer tasks with estimating some of the key circumferences of the body segments, based on the measurement data from both of the front and side view cameras. Instead of using just one camera view, it uses a pair of camera views that are 90 degrees apart to determine dimensional information thus building up a more complete three dimensional picture of the human body using only a camera and a non-3D scanning system.

Front view camera makes lateral width measurements of each body section using the horizontal width of sections like chest, waist, hips, upper arm and thigh looked at head-on. The side view camera, located at a 90 degree angle, takes the "front to back" measurements of the same segments; taking the physical depth and thickness of each body part that cannot be seen from the front view only.

Both of these are complementary measurements taken together and together approximate each body segment as an elliptical cross section, with the width taken front view being the major axis and the depth taken side view being the minor axis. Human body segments can be modelled using a well-known geometric approximation that is an ellipse, which compromises between anatomical realism and simplicity of computation. Importantly the accuracy of the system is based on an approximation of circumference,

also known as girth and not cross-sectional area, of each of the body segments. Clinically and practically important measurements related to fitness, body composition, and garment sizing and monitoring for future use are obtained using girth values.

Girths-derived from this layer include Chest Girth, Waist Girth, Hip Girth, Arm Girth and Thigh Girth. All of these measurements share a common ground to be applied and/or interpreted across sports science, medical and fashion industry, and therefore make NeoFit outputs measurable and comparable to population norms that are directly applicable in real use. As such, this layer is between this two dimensional pose land-mark data and meaningful 3D body measurements without having a depth sensor or volumetric scanning equipment.

4.2 Architecture Components

1. Login and Authentication Module

Login and Authentication Module manages user identity, account security and controlled access to the system. Supabase Authentication will be used as a user management and authentication service for the proposed system. Secure registration, login and access to a user's personal measurement and progress records. At registration, users' credentials are safely recorded and treated by the Supabase authentication framework, which incorporates business standard security measures, for example, encryption of passwords, secure session handling, and so on.

The combination of Supabase Authentication offers a number of benefits, such as the provision of secure user authentication, Account control and concurrency, and easy control of the cloud-based PostgreSQL database. Every individual measurement record is associated with a unique authenticated individual to guarantee data personal privacy and to forestall unauthorised access to individual fitness data. The authentication system is also persistent, allowing for users to login to apps across different devices and not lose their continuity of account. The use of a cloud-based authentication system further increases the security, scalability, and reliability of the system, and also offers a customized experience for each user.

2. Input Capture Module

The Input Capture Module is the initial module that has to capture the real-time visual data of the body that is necessary for the body measurement and hypertrophy analysis. The module uses the standard web camera for capturing images of the user and triggers the whole measurement process. The previous one used to take images of the front and side of the vehicle with manual keyboard controls, but in the new system, an automated multi-view image acquisition process is used. The user is led through a series of structured captures, with three different views automatically captured: Front View, Left View, and Right View. The three such viewpoints provide all visual data on the user's body size and allow for more accurate estimation of girth and analysis of the user's body.

The system constantly tracks the user's position during the viewing process and gives on-the-spot instructions for capturing the required views in the proper way. The front has to be used to take width related body measurements and the left and right views are used to get profile data to help estimate body contours and girth measurements. After successfully taking three views, it automatically gets processed by the next modules, without any manual effort. This automated capturing mechanism makes it easier to use, less work for the user and more consistent across measurement sessions.

The Input Capture Module applies a number of capture constraints to ensure the accuracy and reliability of measurements. The user should be entirely in camera view, have a correct standing posture and be well lit when taking the images. The camera position should not change during the process of capturing; too much movement by the user should be avoided to avoid errors of the landmark detection. These conditions are checked in the system before inputting captured images, and only high-quality input images are input to the pose estimation stage. The body development indicators used for this standardized capture are extremely useful for consistent measurement and for tracking body development over time.

The Input Capture Module automatically captures the body's front, left, and right views, while maintaining stringent quality control measures, offering a solid basis for body measurement, girth estimation, and long-term muscle hypertrophy analysis. Central to

this is the need for reliable, standardized visual data for analysis at the following processing stages, which is also provided by this module.

3. Pose Estimation Module

The Pose Estimation Module is based on the MediaPipe Pose framework that is used to detect human body landmarks in real-time. This module is in charge of determining important anatomy points that are needed to extract measurements. MediaPipe Pose provides:

- 33 body parts identified.
- High-speed real-time processing
- High performance in different environmental conditions. Out of these landmarks, the system will only choose the ones that are to be measured:
- Shoulders
- Elbows
- Hips
- Knees The landmarks are symbolized as:

$$(x, y, z, visibility)$$

where: (x, y) are normalized image coordinates.

Where (z) is relative depth.

- visibility embodies score of confidence.

4. Landmark Extraction Module

The Landmark Extraction Module takes the output of the pose estimation step and selects the necessary keypoints on which additional calculations are to be done. This module does the following operations:

- Choice of appropriate anatomical landmarks.
- Pixel coordinates to normalized coordinates.
- Data organization to compute measurements. For example:

Left Shoulder $\rightarrow ((x_1, y_1))$

Right Shoulder $\rightarrow ((x_2, y_2))$

This module decreases the computational costs and guarantees the efficient downstream processing by isolating only the required landmarks.

5. Measurement Computation Module

This module uses the Euclidean distance formula to compute distances between pairs of selected landmarks. These distances are important body dimensions that are needed in the analysis of hypertrophy. The formula used is:

$$Distance = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Measurements computed include:

Length of arm (Shoulder to Elbow)

Shoulder Width (left shoulder to right shoulder)

- Thigh Length (Knee to Hip)

At this point all measurements are in pixel units and are the quantitative foundation of further processing.

6. Scaling and Normalization Module

The Scaling Module converts pixel-based measurements into real-world units (centimeters), making them meaningful and comparable.

The scaling factor is computed as:

$$cm_per_pixel = \frac{0.93 \times H}{P}$$

where:

(H) is the height of the user in centimeters.

- (P) is the pixel height (head-to-foot distance in image) This normalization ensures:
- Inter-user consistency.
- Comparability across sessions
- No dependence on camera resolution. In the absence of this, measurements would only be relative and cannot be applied to real-world analysis.

7. Pose Validation Module

To enhance reliability in measurements, the system has a Pose Validation Module that determines whether the pose taken is acceptable to be analyzed. Validation rules include: Front View Validation: • Shoulders should be in a horizontal position. • Body should be facing the camera. Side View Validation: • Shoulders should be overlapping the X-axis. • Body should be in a side position. The poses that are invalid are rejected, so that only high quality data is handled. This helps a great deal in minimizing measurement errors due to wrong posture.

8. Multi-View Integration Module

Multi-View Integration Module is a combination of the measurements taken in front and side views to estimate the three-dimensional body structure. • Front View → Gives width data. • Side View → Thickness information. The system also addresses the weakness of single-view systems, which cannot be used to capture depth by combining these two views. This combination allows a more precise estimation of body sizes.

9. Pseudo-3D Modeling Module

This module is an approximation of the size of the muscles. The cross-section of the muscle is assumed to be an ellipse. The area is determined as:

$$Area = \frac{\pi \times width \times thickness}{4}$$

This enables:

- Measurement of muscle size.
- Improved hypertrophy tracking
- More significant comparison than linear measurements. It is one of the fundamental innovations of the system, which enables pseudo-3D analysis with only 2D information.

10. Data Storage Module

The Data Storage Module manages the safe preservation of the calculations made by the system, the data of the users and all the progress which the system gives along the years. The system will have an inherently different approach from the conventional file-based storage, like CSV files, where I propose to use Supabase Cloud Database for the backend storage solution, which runs on PostgreSQL as a relational database. Such a cloud-based architecture provides for structured, scalable, and persistent storage of measurement data, which access is secure and is data managed efficiently. Every measurement session gets added to a database record with a different result, which includes information.

Supabase PostgreSQL offers a number of benefits over local file storage. Records can be stored centrally in the cloud, indicating muscle development and body status over a long period of time through several sessions. A user can go back and fetch measurements from the past, check how his/her progress over time, and visualize his/her progress over time using analytical dashboards. The architecture of the database also enables efficient queries, quick data retrieval, and robust data persistence—limiting the danger of data loss that can come with local storage solutions. Additionally, the cloud-based architecture allows for multi-user support, remote access, and easy integration with web and mobile applications. This allows the Data Storage Module to store measurement data, but also provide the basis for future use in longitudinal studies, for tracking student progress and for extending the system to include elements such as predictive analytics and tailored fitness advice.

11. Visualization Module

The Visualization Module will take the data of saved measurements and fitness data and make it presented graphically so that the user can easily understand how his/her body is developing and how his/her fitness is changing over time. The module fetches data from Supabase cloud PostgreSQL database and visualises it in an interactive charts, trend graph, progress indicators and predictive visualisations. The main parameters displayed by the body measuring system are Torso Girth, Limb Measurements, Body Weight and Body Fat Percentage.

Torso Girth is a collection of circumference taken from various points on the upper torso, e.g., the chest, waist, and abdomen. Girth measurements are important to help users track body composition shifts, muscle development, and fat gain in the torso. Changes in the chest circumference can be an indicator of muscle growth in the upper body segment, whereas changes in the waist circumference can offer information on fat loss/gain. The visualization of the torso girth trends allows the users to follow physical modifications occurring over time that would perhaps not be apparent without conducting a visual inspection.

Body parts (e.g., arms, thighs) are measured using Limb Measurements. The measurements are a reflection of true muscle gain and training result in strength training. The system allows trend graphs of the different limbs to be displayed over time to track muscle growth, highlight periods of rapid muscle growth and assess the success or failure of particular training plans. The historical comparison is also useful when users want to find out if they're progressing in their muscle growth throughout the body or not.

The other parameter to view under the dashboard is Body Weight. Data collected during weight measurement could serve as a tool to gauge body mass or provide information to the user with regard to the impact of exercise and nutrition changes on their physical state. To give the overall complete fitness progress, weight, along with other body dimensions is displayed as well. Comparing the trends in weight gain or loss alongside girth measurements allows users to better identify how weight is being deposited, whether as fat or muscle, or simply body mass changes.

Body Fat Percentage offers a practical indication of the body composition at this level, by looking at what percentage of the body is fat compared to the total body weight. While the significance of weight gain is obvious, body fat percentage can be used to help assess the quality of the physical change. A reduction in body fat percentage and a relatively stable or unchanged in the muscle girth measurements are very likely indicators of fitness improvements. Body fat visualization helps users look back at the results of diet, cardio and resistance exercise regimens.

The Visualization Module provides for two primary types of data analysis: Historical Trends and 16-Week Projections. Previous measurement records are shown as a time-series graph for historical trend analysis to watch the changes of torso girth, limb measurements, weight, and body fat percentage in previous measurements. These visualizations enable the detection of growth patterns, consistency of growth, plateaus and growth improvements. Historical data analysis is useful to understand the impact of training, nutrition and lifestyle decisions on long term fitness.

The system also allows for 16-Week Projection Analysis, which relies on the body development patterns of previous measurements to predict future body development.

The projection module of the system estimates the change in torso girth, limb sizes, body weight and body fat percentage within the coming sixteen weeks. These predictions guide users to establish realistic fitness goals, gauge potential progress, and make informed fitness and nutrition decisions. The Visualization Module uses past measurements and predictive analytics to provide actionable insights from raw measurement records and gives the chance to make decisions that are informed by data and create better fitness plans for tomorrow.

12. Diet Recommendation and Nutrition Module

The Diet Recommendation and Nutrition Module will help users to meet their fitness and body composition requirements by offering individually tailored guidance on nutrition along with body measurement analysis. Incorporating nutrition monitoring further boosts the system's effectiveness, given that diet is a key determinant of muscle size and growth. Collects and manages user-specific information including age, gender, height, weight, activities, fitness, and body composition. This information, in turn, allows the system to calculate the nutritional needs for the day and to make appropriate feeding suggestions.

The module provides valuable nutritional information like calorie, carbohydrate, protein and fat consumption per day. The values are utilized to form privateised meal suggestions that meet to the user's objectives, be they muscle building, weight reduction or maintaining their weight. The nutritional targets and guidelines are represented by the dashboard, so that one can see what nutrients they need to consume and how they are spending them everyday.

The Diet Recommendation and Nutrition Module utilizes body measurement and progress tracking functionalities of the overall system. Your nutritional advice can be used to meet your nutritional needs as you go further on your journey, since new measurements and body composition information can be collected over time. This approach makes dietary guidance able to be followed throughout the fitness journey. The nutrition information and user preferences are stored securely in the Supabase PostgreSQL cloud database, to be tracked and analyzed over the long term.

The system would improve fitness monitoring capabilities by offering a more holistic solution, as it deals with both body measurements and personalized nutrition planning. This combination of dietary guidelines, body composition monitoring, and growth analysis empowers users to make informed decisions about their training and nutrition. This comprehensive approach enhances user engagement, promotes healthier selections and is useful in muscle hypertrophy and fitness management.

If the validation criteria is not met, then the system does not compute the measurement but acts on it as per threshold based validation mechanism. The body measurements extracted during processing, and the detected landmarks, are then checked with known threshold parameters which define the range and anatomical proportion of the proper body. The calculated values will not be accepted if one of the values is over or below the limits of tolerable exposure.

Then, an alert or error message will be created immediately declaring that the measured parameters are outside the bounds of the permissible parameters. This helps the user realize that the current capture is not compliant with the validation based requirements and that the measurement is unable to be processed. The inaccurate or abnormal measurement data is not stored in the database, so that the historical data and future analysis will not be affected.

The alert mechanism informs the users possible problems like wrong posture, wrong body position inside camera frame, you don't see your entire body or excessive distance from the camera, etc.

4.3 Methodology

The procedure proposed system methodology determines the entire process execution flow in which raw visual input is organized that is converted into meaningful muscle growth measurements and analytical understanding. It is structured as a sequential pipeline, which is consistent, accurate, and reproducible between sessions. This is a methodology that uses computer vision, geometric modelling and data processing algorithms to automate the whole workflow, as opposed to the traditional methods which employ manual measurements or subjective interpretation.

The methodology specifically addresses three core challenges inherent in muscle hypertrophy tracking. First, it ensures reliable extraction of body measurements through robust pose estimation and landmark detection. Second, it provides a mechanism to convert pixel-based measurements into real-world units, enabling meaningful comparison across users and sessions. Third, it incorporates a longitudinal tracking framework that allows users to monitor muscle growth trends over time. By integrating these aspects into a unified pipeline, the system achieves a balance between accuracy, usability, and scalability.

4.3.1 Overall System Workflow

The NeoFit body measurement pipeline is a multi-stage process that enables the creation of accurate, reliable, and user-friendly pipeline from authentication to displaying analytics and AI-driven forecasts. All of the above steps rely on the successful completion of the previous steps, thus creating a sequential processing chain that transforms raw data from users into validated body measurement data, which is stored and then visualised.

The workflow starts with the Login stage, where the user logs her/his identity in the NeoFit application. Secure authentication allows all following measurement data to be assigned to the appropriate user account and history records to be private to the account and remain on the back logs between user sessions.

Post successful login, the user then moves to the Profile Setup stage. At this point, the system gathers essential demographic and physical data from the user, such as the user's height, used as the main scalar for all the scale conversions from pixels to centimetres in the following stages. The very first step that gets taken to measure a person is to get the baseline information about the person in this profile.

The system then starts capturing the multi-view sequence. The first step is to do Front View Capture where the user places his or herself in front of the camera directly. Then users will turn themselves 90 degrees to reveal the left side (Left Side Capture). After that, Right Side Capture is taken with the user showing their right profile, thus allowing the cameras to capture the depth from the opponent side, and the resulting images can then be averaged symmetrically to decrease pose induced asymmetry errors.

After obtaining all three views, the images are then fed into the Pose Detection stage and then into the Landmark Extraction stage where certain anatomical landmarks are detected and their corresponding pixel points are stored. In the Measurement Computation phase, raw pixel-based distances are obtained between relevant pair(s) of landmark(s). The Scaling and Normalizing step then use the known height of the user to convert the measurement from pixels to centimetres. Multi-View Fusion optimally integrates the scaled measurements from all three views. The fused values are used in the Pseudo-3D Girth Estimation, with Ramanujan's Perimeter approximation for an ellipse used to get the final values of Girths. Physiological Validation takes care of checking the value of all the computed measurements within the anatomically realistic limits before committing the session to Session Storage.

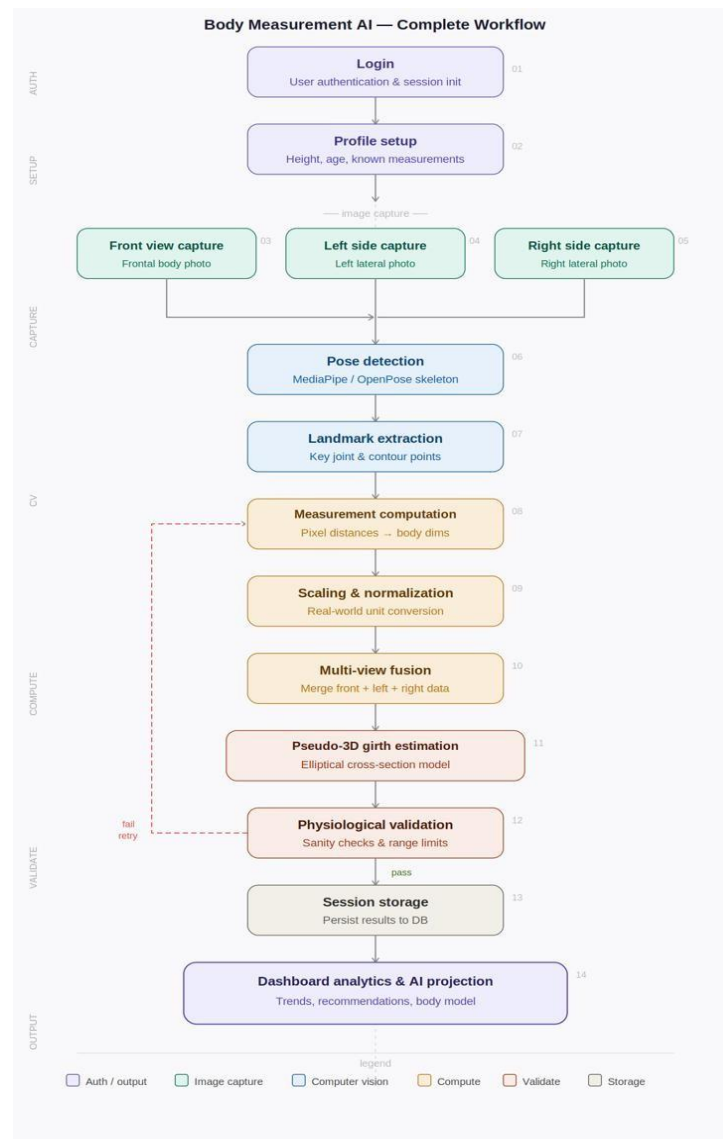


Figure 4. 2 Body measurement workflow

4.3.2 Login and Authentication

The first step in NeoFit workflow is user authentication. After the app is launched, the user will get a login screen or window which will request him/her to enter the credentials that have been registered earlier. If the process of verification is successful, a secure session is created and the recorded and captured body measurement information for that session is properly linked to the authenticated user. Authentication also comes into effect with regards to data privacy as it prevents unauthorised access to past measurement records. When the user accesses the system for the first time he/she is asked to register his/her account and then access the profile initialisation stage.

4.3.3 Profile Initialization

After signing up, any new signing user or user who wants to upgrade the "physical profile" goes through the Profile Initialization phase. This stage gathers a key anthropometric metadata, at least the standing height of the user in centimetres. This height value is used as a reference, the actual true height (scalar) so that all of the height measurement values obtained from images made by the camera are translated to real world centimetre units. Further, the profile can gather extra details like fitness targets, gender, age, etc., which will aid individualize the analytics experience on the dashboard and also feed into the AI projection models. All computations of measurements downstream should then be considered as valid only on the basis of an accurate profile initialization.

You can capture multiple pictures in a single row, as a front, left side, or right side. Capture multiple pictures in a single row; front, left side, right side.

The capture stage is the main data collection step of the NeoFit pipeline. The system automatically leads the user through a structured sequences of three views (S3V) to capture the needed multi-perspective image content to estimate girth size pseudo-3D. In this Front View Capture the user stands towards the camera with arms slightly abducted from the body providing unobstructed view of waist and hip areas. This perspective gives all the width measurements laterally (to and fro) over the whole body at landmarks. The user turns 90 degrees counter clockwise to allow the computer to show the left profile which allows anterior to posterior depths to be taken from the left. In the Right Side Capture, the user turns his or her body another 180 degrees to capture the right side,

giving depth measurements from the other side. By obtaining the depth value from both sides of the person, then the system can take the average of both sides, which helps to compensate for any slight postural or stance imbalance that would normally influence depth measurement taken from just a single side view of the person. The three acquired images from different perspectives are sufficient to cover the multi-view needed for landmark extraction and girth estimation in the next step.

4.3.5 Pose Detection

After obtaining three-view images from the camera, they are filed separately by the pose detection module. Pose detection uses a pre-trained human pose estimation model that localises the human body in each image frame and provides an outline of the human body. The model determines corresponding joint positions in the entire body frame as well as position of any desired anatomical reference points in pixel coordinates. Pose detection runs on each of the three views in turn, yielding a different set of landmark coordinates for each view from the front, left and right perspective images. Therefore, the reliability of all subsequent subsequent extraction of measurements depends on the reliability of the output of the pose detection; inaccuracies in pose detection output can result in the prompt for re-capture if the confidence of the landmarks that are detected is not above a minimum set by the system.

4.3.6 Landmark Extraction

In the Pose Detection stage, pose detection is followed by an extraction stage that identifies And records the corresponding subset of the anatomical landmarks that are relevant to body girth measurement. The detected keypoints for a specific view are known as the shoulders, chest level, waist, hips, elbows, wrists, knees and ankles. For each view, keypoints are detected from a set of defined keypoints that include shoulders, chest level, waist, hips, elbows, wrists, knees and ankles. These known landmarks are subsequently identified by pixel coordinates, and stored in a structured form by body segment and view perspective. The front view is used to get width relevant landmark pairs; the side views are used to get depth relevant landmark pairs. This stage yields a world and result of a highly important set of data that is used as geometric input for the measurement computation stage.

4.3.7 Measurement Extraction

During this step, the system calculates the raw distance between different landmark pairs for every body part in the three different views: Measurement Extraction. The horizontal distance (in pixels) from the left terminus landmark of every segment to the right terminus landmark is calculated from the front view for each width measurement. The anterior to posterior distance of the images is calculated from the side view images at the side of the anterior and posterior boundary landmarks for each depth measurement. These pixel distances are the apparent widths and depths of each body segment as visible in the two-dimensional image plane, which form the basis for real-world measures after scaling and are the geometric basis.

4.3.8 Scaling and Normalization

The raw pixel readings obtained at the previous step doesn't have a natural unit of measurement in the real world and relies on the camera distance and the subject's position. The Scaling & Normalisation stage addresses this by using a person specific Pixel-to-Centimetres conversion factor, based on the user's height that is known. The system calculates the total height of the body that was detected in the image as the height from crown to heel in the image and obtains the user's real world height in centimetres, and the conversion scalar is made by dividing these two numbers. This scalar is then multiplied by all width and depth data for each body segment which is pixel-based to yield measurements in centimetres. To normalize the measurement pipeline and produce the same and comparable outputs for various distances of capture, different zoom levels and frames of varying sizes of the same subject.

4.3.9 Multi-View Fusion

The Multi-View Fusion stage combines measurements independently acquired from the front, left and right views across each of the body segments into a single set of measurements for the body segment. The width obtained from the frontal view is directly used as the width values for each segment. The depth taken from the left side view plus the right side view in each segment is averaged to produce one depth for each of the segments against which the minor difference in postures or weight distribution left and right is minimized. The results and output from this stage are a single table of pairs of

widths and depths for each anatomical body region in centimetre units which are suitable for use as input data for the girth estimation model.

4.3.10 Pseudo-3D Girth Estimation

The model for pseudo-3D Girth Estimation stage, calculates the circumference girth of each body segment using the cross section model of an ellipse, based on the fused width and depth data. The width is considered to be the major diameter and the depth is considered to be the minor diameter of each segment ellipse that approximates the cross section shape of the body at that point. The semi-axes are found by taking half of each side and Ramanujan's approximation formula is used to work out an estimate of the perimeter of the ellipse, hence giving the girth in centimetres. This process is carried out for each target segment individually to yield the final Chest Girth, Waist Girth, Hip Girth, Arm Girth and Thigh Girth values, which are the main output measures of this NeoFit pipeline.

4.3.11 Physiological Validation

All computations of girth measurements are then Physiologically Validated prior to committing any session data to persistent storage. The system takes the measurement of each girth and compares it with a predetermined set of anatomical or physiological thresholds based upon known human physiology and anthropometric data. These are just the range of values between which it is plausible to expect that girths measured in each body part will fall, given the intended population of customers. When one of the values for any girth level does not adhere to one of the girth levels, the entire session is rejected for storage. The user is notified of an error and is asked to retake the session under better conditions if the recorded values were outside of the physiological range. This validation gate allows corrupted measurements caused by poor lighting, occluded landmarks, large posture errors or failures of a model to be thrown away so that the user's historical measurement record is not polluted with such false data or the AI generated prediction of their future posture is distorted.

4.3.12 Session Storage

In the Session Storage stage, successful physiological validation of measurement sessions will be stored for later in the database. Every session record is stored with all the validated Girth measurements validated during the session (Chest Girth, Waist Girth, Hip Girth, Arm Girth, Thigh Girth) and information about the session itself (session id, session timestamp, any session capture condition flags). It is a schema optimized for longitudinal queries to support the dashboard analytics and AI projection modules that will retrieve historical sessions and provide temporal trend comparisons. The system's validation and structure of measurement sessions form the basis for meaningful fitness progression and personalised, AI-driven body composition projections.

4.3.13 Dashboard Analytics and AI Projection

The last step in the NeoFit workflow is providing the processed measurement information to the user via Dashboard Analytics and AI Projection. This displays the last calculated girth calculations and will also show trend graphs for past sessions for each body segment measurement so visualising how each measurement has changed over the last visit. This longitudinal advantage lets users see the effects of their training and nutrition regimes on particular sections of the body, over time. The AI Projection module then - based on the history of each measurement, makes a forward looking prediction about how the user's body dimensions will change as a result of his/her fitness program continuing. These projections take advantage of imperfect cross-correlation, supplementing the information logged by the app with personalized, data-driven advice and recommendations tailored to an individual's specific fitness goals.

4.4 Mathematical Modeling

The analytical foundation that allows the conversion of raw visual data into structured and meaningful muscle measurements is the mathematical modelling of the proposed system that will make this possible. Many mathematical formulations are necessary since the system works with the input of 2D images to translate the pixel-based representations into the real-world units and to estimate the three dimensional muscle structures. This section will give the mathematical basis applied to coordinate transformation, distance computation, scaling, geometric modeling and temporal growth analysis. The main aim of

this modeling is to make sure that measurements in the image space are accurate as well as comparative across sessions. Using clearly defined mathematical relationships, the system becomes consistent, less ambiguous and allows reliable hypertrophy tracking.

4.4.1 Coordinate Representation

Everybody landmark found is in normalized coordinate form as:

$$L_i = (x_i, y_i)$$

where: x_i : The horizontal coordinate. y_i : is the vertical coordinate. These are the coordinates which are obtained in normalized form of the pose estimation model, which is usually between 0 and 1. These normalized values should be converted to pixel positions depending on the size of the image frame to compute measurements. The conversion is done as:

$$x_{pixel} = x_i \times W$$

$$y_{pixel} = y_i \times H$$

where: W = width of image frame. Where: H is the height of the image frame. This transformation makes all landmarks come up in a uniform coordinate system that is appropriate when calculating distances.

4.4.2 Distance Measurement Model

The system calculates the distances between landmarks based on the Euclidean distance formula. For two points:

$$P_1(x_1, y_1), \quad P_2(x_2, y_2)$$

the distance between them is given by:

$$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

This equation gives the line distance between two points in the image plane. This model is used to calculate important body measurements in the framework of the system:

- Arm Length:

$$D_{arm} = distance(shoulder, elbow)$$

- Shoulder Width:

$$D_{shoulder} = distance(left\ shoulder, right\ shoulder)$$

- Thigh Length:

$$D_{thigh} = distance(hip, knee)$$

All the calculated distances here are expressed in pixels and are the base measurements in doing the additional processing.

4.4.3 Scaling Model (Pixel to Real-World Conversion)

The fact that pixel-based measurements are not directly applicable in the real-world context means that there is a need to have a scaling mechanism that can translate them into more standard units like centimeters. This is accomplished by a scaling factor which is calculated based on the height of the user. The scaling factor is given by:

$$Scale = \frac{0.93 \times H_{user}}{H_{pixel}}$$

where:

- (H_{user}) is the user's height in centimeters
- (H_{pixel}) is the height of the body in pixels within the image

The final measurement in centimeters is obtained as:

$$Measurement_{cm} = Measurement_{pixel} \times Scale$$

The constant 0.93 is added to provide a calibration constant to take into consideration practical variations, including inaccuracies in landmark detection, partial visibility of the body and slight distortion in perspective. This empirical correction increases the real-world approximations.

4.4.4 Multi-View Geometric Modeling

The proposed model uses multi-view measurements to overcome the constraints of single-view systems. Two independent dimensions are derived:

- Width ((W)) front view.
- Thickness ((T)) in the side view.

The two measurements are orthogonal dimensions of the body and have been used to estimate three-dimensional structures. The system measures both width and thickness allowing it to record lateral and depth information, allowing a more detailed representation of muscle geometry.

4.4.5 Pseudo-3D Girth Estimation Model

The NeoFit system uses a mathematical precise formula for the perimeter of an ellipse to estimate the circumference of important body parts by roughly approximating each part as an ellipse and using the formula. The way this works is that the system derives clinically relevant measurements of girth from only two-dimensional images taken from orthogonal camera viewpoints, and without using depth sensors, LiDAR hardware or structured light scanning.

Two input measurements from the dual-view capture pipeline are used to specify the geometric model. Variable W is the width of the body segment as seen from the front image and represents the horizontal distance between the furthest end points of the body-segment from the image in the front view at the landmark location. The variable D indicates the depth of the body segment at the side view image and thus the thickness of the segment in its anterior to posterior direction, at the same anatomical position. The sum of the two determines the size of an ellipse that best fits the shape of the body segment in cross section at the position.

The semi-axes of the ellipse are determined from the measurements above. Half of the width ($W/2$) is known as the semi-major axis (a). The semi-minor axis b is half of the overall

depth, or $D/2$. These semi-axes completely describe the elliptical model of the cross-section of the body segment.

Use these replacements throughout the entire report, including the body text, headings, figure captions, table headers, and figure titles (e.g., Figure 2.1).

It automatically replaces headings, body text, figure captions, table titles, table headers and figure titles (like Figure 2.1). It is not necessary for NeoFit to calculate the cross-sectional area anywhere along the pipeline. The system calculates values of the circumferential girths based on Ramanujan's approximation to a ellipse's perimeter and all names throughout the report should be consistent with this.

4.4.6 Temporal Growth Modeling

To track muscle growth over time, the system models measurements as a time series:

$$M(t) = \{m_1, m_2, m_3, \dots, m_n\}$$

where:

- (m_i) represents the measurement at time (t_i)

This representation allows the system to analyze trends and monitor progression across multiple sessions.

A simple linear growth model can be expressed as: $M(t) = mt + c$

where:

- (m) represents the growth rate
- (c) represents the initial measurement

This model enables estimation of future growth trends and provides insights into the effectiveness of training programs.

4.4.7 Error Consideration

The errors that could be created during measurement can be attributed to multiple factors, such as inaccurate pose estimation, camera angle, bad posture, and environmental factors, namely, lighting. In order to measure the error, the system contrasts its measurements with the manual measurements with the equation:

$$Error\% = \frac{|Manual - System|}{Manual} \times 100$$

This measure gives an idea about the correctness of the system and assists in finding the possibilities of improvement.

4.4.8 Model Assumptions

The mathematical model is based on several assumptions that simplify the computation process:

- Body segments are treated as rigid structures
- Muscle cross-sections can be approximated as elliptical shapes
- Camera perspective distortion is minimal
- The user maintains consistent posture across sessions

These assumptions enable efficient computation while maintaining a reasonable level of accuracy for practical applications.

4.4.9 Advantages of Mathematical Model

The mathematical model proposed has a number of benefits. It is computationally efficient and simple and thus it can be used in real-time. Geometric approximations give meaningful representations of muscle size, but do not necessarily involve complex 3D reconstruction methods. The model is also longitudinal analysis friendly as it allows uniform measurement across sessions.

4.5 Validation Logic

The effectiveness of the suggested system will be largely determined by the accuracy of the posture of the user when capturing the image. As the whole measurement pipeline depends on the spatial similarity of identified landmarks, postural variations or camera positioning could cause considerable errors. To solve this, the system has a validation mechanism that provides a quality control layer prior to the calculation of measurements. This validation will make sure that reliable and consistent inputs are only processed and this will enhance the overall accuracy and strength of the system. The necessity of validation is due to the nature of variability of the real-life conditions. In the absence of validation, the system can give inaccurate measurements because of improper body positioning, partial obscuration of body parts, camera distortion or because of movement of the user during the capture. These can skew landmark positions and give inaccurate distance calculations. A validation mechanism will greatly reduce noise and provide consistency across sessions by excluding such invalid inputs. In the front view, the system checks to ensure that the user is facing the camera correctly and in an upright position. This is done by shoulder alignment check which involves the difference between the left and right shoulder in the vertical direction being limited within a narrow range. Mathematically, this can be represented as:

$$|y_{left_shoulder} - y_{right_shoulder}| < \epsilon$$

This is a condition that ensures that the shoulders are at the same horizontal position meaning a stable and upright position. Moreover, body orientation examination is done to ensure that both shoulders remain visible and that there is a distance between them that is greater than a set limit. When these requirements are met the pose is counted as a valid front view; otherwise the pose will be rejected. In a side view, the system is used to make sure that the user has rotated enough to give a lateral view. This is confirmed with a shoulder overlap check, in which the difference between the horizontal position of the left and right shoulders is limited by a small range:

$$|x_{left_shoulder} - x_{right_shoulder}| < \delta$$

4.5.1 Need for Validation

Without system validation, our measurements could be misleading because of the following factors. Computer vision is highly dependent on its input and any inconsistency in body pose or environmental conditions may influence the landmarks.

A common problem is the incorrect posture of the user, who does not keep a straight posture. This may result in misplacement of landmarks, which will affect distance estimates. Likewise, camera positioning, especially when body parts are partially obscured (e.g., shoulders hidden, legs hidden), may make the Kinect system unable to detect body points.

Camera positioning errors, where the camera is not placed at the right angle, is another issue. This may lead to perspective distortion and measurement errors. Also, user movement and instability can affect landmark detection, which can result in unreliable readings.

Given these factors, validation is crucial. It serves as a quality check to ensure only the most consistent and accurate data are measured, thus increasing the accuracy of these measurements.

4.5.2 Front View Validation

When capturing front views, the system needs to check that the user's head and body are correctly orientated towards the camera. This is required as front view measurements like shoulder width are only possible if posture and orientation is correct.

The first condition, the shoulder alignment, makes sure that the shoulders are in a horizontal line. This is represented as:

$$|y_{left_shoulder} - y_{right_shoulder}| < \epsilon$$

This ensures the difference in height between the left and right shoulder coordinates is minimal. If this difference is small, it means that the user is not tilting his/her head. Good alignment is crucial for estimating the width.

The second check is to confirm that the user is facing the camera. Here, the user's two shoulders need to be visible and the distance between their centers needs to be greater than a threshold. This ensures that the pose is facing the camera and not turned at an angle or to one side.

If these conditions hold, the posture is deemed to be a front view. A failure of either condition implies the capture is invalid, and so the user is asked to adjust their pose.

4.5.3 Side View Validation

When capturing the side view, the system needs to check that the user has turned enough to be seen from the side. This is necessary as the side view is used to capture thickness, which cannot be determined from the front view.

The main validation criteria is the shoulder overlap check:

$$|x_{left_shoulder} - x_{right_shoulder}| < \delta$$

It guarantees that the absolute difference of horizontal coordinates of the shoulders is small. With the user standing in a proper side pose, the left and right shoulders are practically on the same X-axis, as the face of the user is turned to the side.

The second condition is the decrease of the shoulder width. The shoulder width (measured in a two-dimensional image) should be much less in a side view than in the front view. This shows that the person has rotated and is no longer looking at the camera.

If these two criteria are met, the pose is considered a side view. If not, it is rejected and the user is instructed to change pose again.

4.5.4 Validation Thresholds

The success of the validation depends on threshold values, embedded in the validation process, which control the variations in alignment (denoted ϵ and δ). These parameters are set empirically from experiments and system results.

It's important to select suitable threshold parameters, as this determines the trade-off between sensitivity and user-friendliness. If the thresholds are too stringent, poses may be rejected often and it may be challenging for the user to capture the required poses. However, if the thresholds are too relaxed, the system will not reject invalid poses and measurements will be inaccurate.

As such, the thresholds are set to achieve a balance. They provide enough flexibility for different poses while still constraining the user to ensure the accuracy of the measurements. This makes system operation safe and convenient.

4.5.5 Handling Invalid Inputs

The NeoFit system features a special physiological validation target to ensure that measurement data is stored intact and AI-derived analysis results are accurate. Before any session is written to the database, all body girders that have been rebuilt in the session are passed through an extensive check. Each girth measurement (Chest Girth, Waist Girth, Hip Girth, Arm Girth, and Thigh Girth) is evaluated independently against set thresholds of anatomical measurements that encompass the biologically reasonable range of the girth measurement in question for the user populations anticipated for the device. Anthropometric thresholds have been derived from "normal anthropometric measurements in healthy individuals", and are designed to detect measurements which are physiologically unreasonable because of the typical proportions of a healthy human body.

When a single measurement in a session is above the upper or lower limit of a threshold, that session will be rejected and is not saved in the database. The all-or-nothing rejection policy guarantees that a session can only pass through the rejection filter in full or not at all, which will not affect subsequent sessions either in terms of longitudinal trend analysis, or in terms of generating skewed AI projections.

Then, if the session is rejected, an error message is created which is shown right on the user interface of the application. In this notification, the physiological limits outside of which the captured measurement(s) have taken place is clearly communicated to the user and encourages him or her to repeat the capture process with the following requirements being met: full body visible from the camera, proper posture and adequate lighting.

It is a very important validation check within the pipeline process. As AI-driven body composition projections and longitudinal tracking can be domains of AI, it allows deserving clinical information to rise again in the historical measurement history of each user, while not allowing an unrealistic measure to affect the entire statistical information regarding the measurement within the workflow.

4.5.6 Importance of Validation

Validations are plays a significant role in the system's success. It provides consistency for measurements by controlling input variables. This is particularly important for longitudinal measures, where comparisons between sessions are required. It also helps to minimise errors by preventing incorrect data from being processed. This leads to more precise measurements and increases the system's trustworthiness. In addition, the validation process aids in robustness, enabling the system to function under real-life conditions with diverse poses and environments. The system continues to function correctly even with suboptimal data, due to the use of high-quality measurements.

Overall, validation is an essential feature that improves the accuracy, precision, stability and reliability of the system, ensuring it can be used in realworld scenarios for tracking muscle hypertrophy.

4.6 Design Advantages

The AI-Based Landmark-Driven Muscle Hypertrophy Tracking System has several key benefits over both body measurement techniques and current digital fitness tracking systems. All these benefits are realised in the same framework, by combining computer vision, pose estimation, multi-view body analysis, girth estimation, cloud database management, and tracking of the longitudinal growth.

One of the major advantages of the system is the complete automation of the measurement process. Traditional body measurement methods involve extensive use of measuring tapes and manual operation, including inconsistent placement of the tape, body posture, body measurement method and human judgment of the data. However, the proposed system automatically extracts the anatomical landmarks using MediaPipe Pose and calculates the body measurements without needing manual measurements. This automation minimizes the impact of human error, boosts repeatability, and guarantees measurements stay consistent throughout multiple sessions. This means that body measurements can be made in an objective and accurate manner without the need for another person to help the user.

One of the other key benefits is the adoption of a multi-view body measurement system. The proposed system is an image-based system that differs from conventional systems in that body information is captured from 3 different angles (Front (F), Left (L), and Right (R)). The front view gives information about the width of the body, the left and right views give information on the profile of the body, its thickness and contour. These three measurements can be used together to create a more comprehensive view of the human body. This multi-view approach is able to reduce the restrictions of single-view systems and enhance the accuracy of body dimension estimation.

One of the major advantages of the system is its girth estimation feature. For fitness and anthropometric measurements, arm girth, chest girth, waist girth, and thigh girth are commonly used measures of muscle development and body composition. The proposed system uses geometric relationships based on measurements taken from a number of different views to estimate these girth values. Girth measurements are also very similar to standard fitness evaluation procedures, and therefore are useful for users to track hypertrophy and physical development in a more realistic manner. This enables users to assess muscle growth to a more effective level, rather than just on linear parameters.

The system also has real time processing capabilities. Optimized computer vision and pose estimation algorithms allow for instant detection of body landmarks and measurement calculations in seconds following capturing a picture. Immediate processing ensures users receive instant feedback and minimizes delays, improving their experience. Therefore, the

system can be used in gyms, fitness centers, rehabilitation centers, and personal fitness applications where a swift measurement result is desired.

Another significant benefit of the suggested system is growth monitoring in the longitudinal direction. The system keeps a record of the measurement over a long time period, unlike the traditional measurement methods which only give the measurement value at a particular point. Measurement data is stored safely in a Supabase Cloud Database running on PostgreSQL. This way users can see how their body is changing over weeks, months or years. Historical measurements can be displayed in graphs or progress reports, aiding users in recognizing development patterns, proficiency in training, and long-term fitness trends. This makes the system more than a mere measurement instrument, it becomes a thorough fitness analytics platform.

The cloud-based database architecture also contributes to the scalability and reliability. Supabase PostgreSQL offers centralized storage, secure management of data, and efficient retrieval of user records. Cloud-based storage can help to avoid data loss and allow users to access their data from other devices and locations, unlike local storage. Additionally, the database structure is designed to be expandable for future use, allowing for easy additions to the system as the number of user increases and more data is added.

Another benefit of the proposed system is its cost-effectiveness. There are many advanced body measurement technologies that require the use of a 3D scanner, depth sensor or specialized hardware, which are costly. The solution proposed, on the other hand, involves a standard webcam/smartphone camera, which is much cheaper and more readily available. This relatively inexpensive application helps make it more likely for fitness enthusiasts, trainers, schools and health care professionals to adopt it.

Validation mechanisms are put in place to increase measurement reliability. The system checks for body posture, landmark visibility, and image quality prior to accepting the measurements. Poor poses, poor coverage of body parts and poor quality images are rejected automatically. This procedure helps to avoid erroneous measurements and guarantee that only good data is analyzed. This results in better reliability and trustworthiness of the generated results.

The modular and extensible system is another great benefit. The image acquisition, pose estimation, measurement extraction, girth estimation, database management and visualization modules are independent. The modular format makes for easy upkeep and makes future upgrades easier without significant re-architecting. Future enhancements may involve AI-driven workout suggestions, forecasting muscle development, custom training routines, integration with fitness gadgets, and advanced analytics dashboards.

In summary, the proposed system is a comprehensive and practical solution for muscle hypertrophy monitoring, thanks to its features and capabilities. The system overcomes the main drawbacks from the traditional measurement methods, while introducing a scalable, accessible and easy-to-use platform for modern fitness tracking and digital anthropometric analysis.

5 Results and Discussions

Preamble to the Chapter

This chapter is the experimental test of the suggested AI-based muscle hypertrophy tracking system and the results of the test. This chapter will evaluate the reliability of the system to measure body parts, to be consistent in different capture sessions and to give meaningful pseudo-3D estimates of muscle growth. The test is based on the real-life performance of the system in controlled conditions, with special consideration of measuring accuracy, the strength of pose validation, and the utility of the resulting growth trends. The results are evaluated not only according to the numerical output but also according to the usability and reliability. As the proposed system is a fitness tracking and progress monitoring system, not a medical-grade diagnostic instrument, the evaluation will focus on repeatability, consistency of trends, and approximate accuracy. It also has a comparison with the manual measurements made using a measuring tape which will provide a pragmatic reference point on how the system is performing.

5.1 Experimental Setup

The proposed system was tested using a standard webcam based system in controlled environment. It was designed so that the testing environment is representative of a realistic user scenario in a sufficiently stable environment to assess the performance of the system with a certain degree of accuracy. The three main parameters assessed were correctness of measurement extraction, consistency of measurements between sessions and the reliability of pseudo-3D muscle size estimation. The visual input device used in the test setup was the inbuilt webcam of a laptop. The video capturing and processing were done using OpenCV, and the real-time landmark detection and tracking were done using MediaPipe Pose.

The goal of this configuration was to make it simple and easy to use, without the need for specialized hardware, but have a strong cloud based backend infrastructure. Data collection process was carefully designed to be consistent in all sessions. The first one was the user's height, which had to be entered to convert the measurements from pixels into

real world units. It was then taken a front view image using the F key and the side view image using the S key. Both views were then processed, and the user confirmed and saved the measurements on the C key. All the measurements and analysis results generated were automatically stored in the Supabase PostgreSQL database once they were confirmed.

The system was used several times over time and so could not be used as a one-off measurement instrument, but as a longitudinal system. In order to check the accuracy, the measurements obtained by the system were compared with the measurements obtained manually by a measuring tape. This comparison was useful as a benchmark to determine the difference between automated and manual methods. Manual validation was found to be necessary since it allowed to evaluate the results generated from the model with respect to their applicability in the real world environment as well as consistency of generated results within the model.

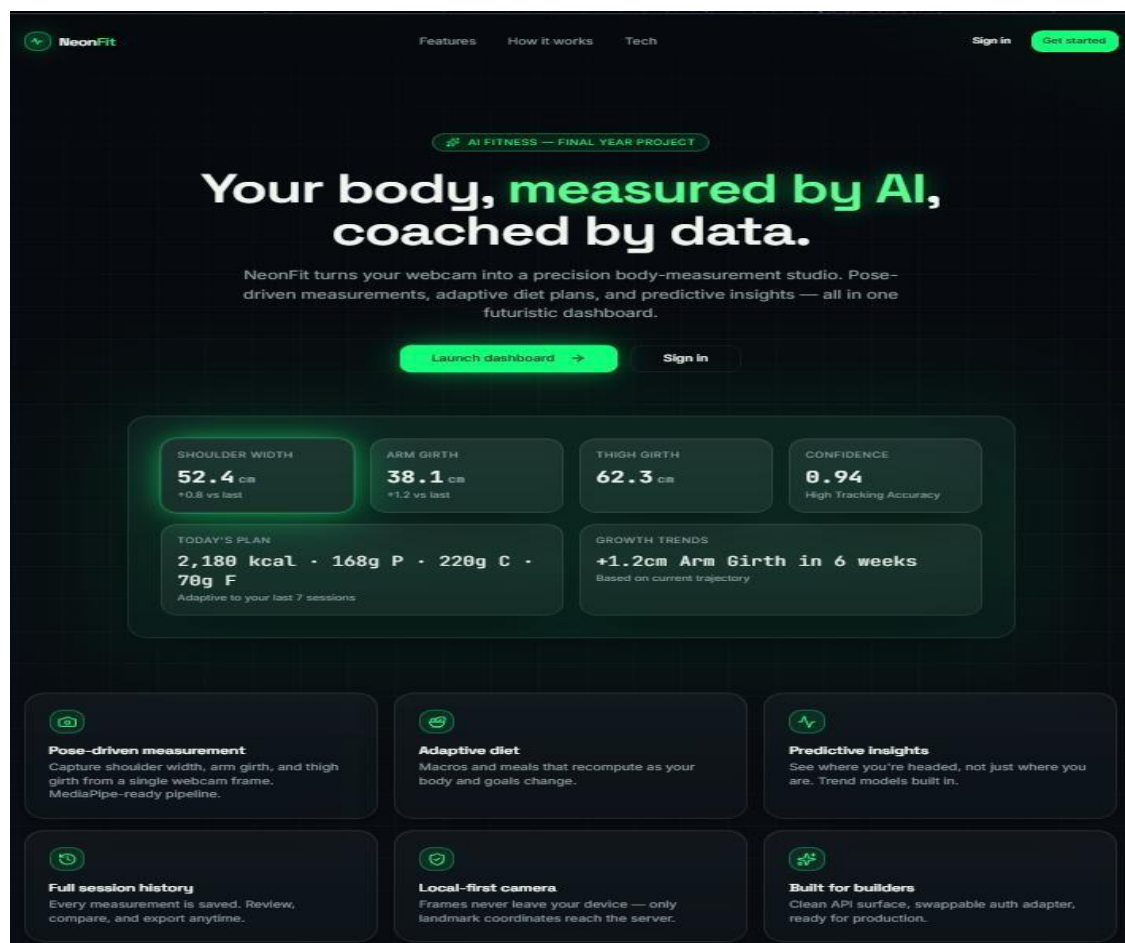


Figure 5. 1 Landing Page

5.2 Performance Evaluation

Three key metrics—measurement accuracy, consistency, and practical usability—were used to assess the performance of the AI-Based Landmark-Driven Muscle Hypertrophy Tracking System. These parameters have been chosen since they directly depend on the ability of the system to track muscle hypertrophy reliably.

The accuracy of the values produced by the measurement system relative to the manual measurement is called the accuracy of the measurement. The experimental results indicated that the measured values of the body dimensions of arm length, shoulder width and thigh length were very close to the manually measured values, with an average error of approximately 2–3%. This accuracy is deemed to be adequate for fitness monitoring and progress tracking applications. Consistency is a test that assesses how repeatable the system is, that is, how similar its output will be if measurements are repeated under the same conditions.

The test showed that there were only small differences between the results of repeated tests, which suggest that the pose validation mechanism and scaling model have stable and repeatable results. This uniformity is important because hypertrophy tracking only relies on the changes that occur within the body and not on measurements that change as a result of the measurement. Practical usability is the ability of the system to be used in practical environments. It requires no expensive hardware or specialized scanning equipment; only a standard webcam, OpenCV and MediaPipe Pose are needed.

The system is simple, easy and convenient to use since the user is required to only provide the height and capture two images, one of the front and one of the side of the body. The system collected the following parameters during evaluation: arm length, shoulder width, thigh length, arm area, thigh area, and the time. The physical body size (linear measurements) is different from the pseudo-3D geometric modeling of the arm area and thigh area which estimates the size of muscles. The timestamp will also enable these measurements to be stored and compared over multiple sessions to monitor muscle growth and development over time. Overall, the performance evaluation demonstrated

that the system achieves a good balance between accuracy, consistency, and usability, making it a practical solution for automated muscle hypertrophy tracking.

This sample output shows that although linear measurements like arm length, shoulder width, and thigh length are being recorded by the system the pseudo-3D values like arm area and thigh area have also been derived. The timestamps will enable the data to be longitudinally tracked since the measurements of the various sessions can be compared by the system and the growth can be observed over time. Comparison between manual measurements and system measurements is summed up below:

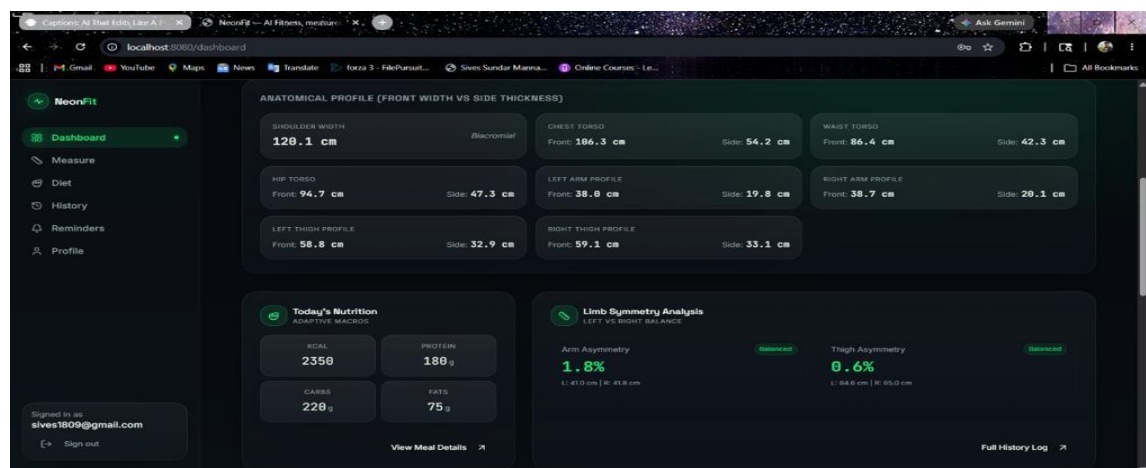


Figure 5. 2 Dashboard 2

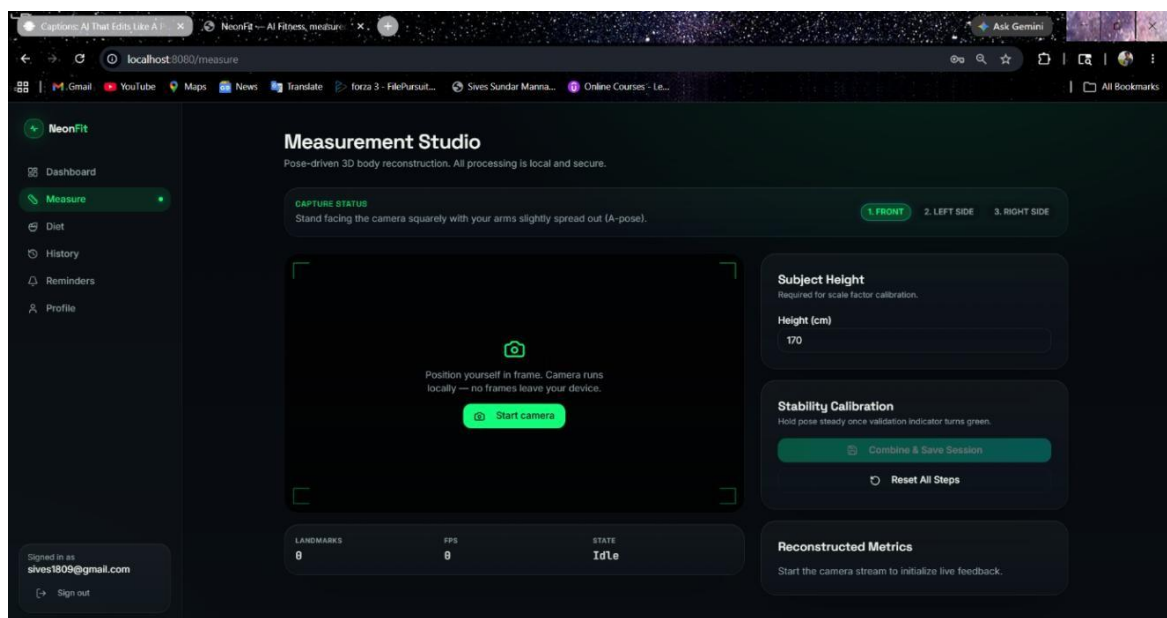


Figure 5. 3 Measurement Studio

The table shows that the system yields values which are quite close to the manual values. The error is also minimal and is normally in the range of about 2-3 percent. Such a degree of deviation is minimal enough to be reasonable as a means of fitness tracking, particularly when its primary goal is to track relative developmental patterns and not to offer medical quality accuracy. It is also found that the system is reproducible among various sessions. The variations in the repeated measurements are small implying that the pose validation module and scaling mechanism are performing well. This uniformity is essential since tracking of hypertrophy relies on observing changes over a period of time; in case the system itself generates erratic values, then the growth patterns would not be significant. In general, the evaluation of performance shows that the system has a viable balance between usability and accuracy. The scores are similar to the manual scores, and the system is stable to serve as a repeated progress monitoring system.

5.3 Key Observations

The AI Based Landmark Driven Muscle Hypertrophy Tracking System dashboard offers a few key metrics which enable users to keep track of body size, muscle symmetry, nutrition and general body development. The Anatomical Profile section shows the body characteristics extracted from the front and side views by landmark detection and geometric modelling. The width of the shoulders, chest, waist, top torso, left and right arm profile and left and right thigh profile are physical measurements of various body regions. The distance between the anatomical landmarks detected by the pose estimation model is used to calculate these measurements, which are then scaled to real world centimeters with a scaling mechanism. These values indicate the percentage changes in body structure and muscle development over time.

Shoulder Width is the width of the shoulders and is an important measurement of upper body development between left and right shoulder landmarks. The width of the shoulders is another indicator of the development of the deltoid muscles, and helps create a wider body. Chest Width is a parameter that represents the width of the chest area, which can be used to assess the development of the pectoral muscles.

Further, the Waist Width parameter represents the width of the waistband, which can be helpful for studying the body proportions and fitness growth. The Top Torso Width measurement gives information on upper torso development, and arm and thigh profile measurements are the thickness and size of the limbs from various angles. These measurements make up an overall anatomical image of the user.

The dashboard also has a Limb Symmetry Analysis section, which checks the balance between the left and right sides of the body. Measurements of the corresponding limbs are compared to determine parameters like Arm Asymmetry and Thigh Asymmetry. The values are the percentage difference between the left and right sides. Higher asymmetry percentages can be indicative of unequal development or training and lower percentages will mean that there is better muscular balance. For athletes and fitness enthusiasts, this analysis can be of great value as it helps prevent injuries arising from imbalanced muscle development and enhances overall physical performance through balanced muscle growth. The system automatically identifies if the symmetry status is normal or if it needs to be attended.

Today's Nutrition is another important part that shows nutritional information like Calories (KCAL), Proteins, Carbohydrates, and Fats. Calories are the amount of energy you need to keep your body working and to build muscle. Protein is crucial to muscle repair and hypertrophy as it helps in producing amino acids that are required for the development of muscle tissue. Carbohydrates provide the main source of energy for exercise and during everyday life and fats are used for hormone production and health. The system combines nutrition data with body measurements, enabling users to grasp the link between their diet and the formation of muscles and their physical growth. This helps users to make better-informed choices about fitness and nutrition plans.

The overall purpose of these parameters is to transform raw image data into meaningful fitness insights. The system utilizes computer vision and pose estimation, landmark identification, scaling models and pseudo-3D geometric analysis to achieve objective body measurements and the monitoring of hypertrophy trends.

The dashboard allows for a long-term analysis of muscle growth, body symmetry and nutritional efficiency through the storage of measurements across sessions. This gives the users a scientific and data-driven method to track fitness progress without having to resort to manual measurements or visual comparisons.

5.4 Discussion

The results of the experiment indicate that the proposed system is effective in overcoming some of the drawbacks inherent to conventional muscle tracking systems. Automated landmark extraction, multi-view capture, geometric modeling, and longitudinal storage make up a unified pipeline yielding valuable and interpretable measurements. Specifically, the system demonstrates the possibility to create a viable hypertrophy tracking system that can be based solely on regular imaging devices and pose-estimation technology. Automated extraction of measurements is one of the significant successes of the system. In contrast to manual techniques that require user dexterity, tape positioning and subjective analysis, the proposed system extracts measurements in a computational manner.

This minimizes the human error and increases repeatability of the process. The fact that the deviation is low when compared to that of manual measurements is an indication that the system is precise enough to track progress as time goes by. Real-time processing is another important accomplishment. MediaPipe Pose and OpenCV enable the system to be used in an interactive mode, which is significant to a user-facing application. Real-time feedback is particularly handy in the capture mode since it assists the user to stay in a proper position and enhance the quality of input data. This directly helps the validation logic and helps in enhancing measurement reliability.

The visualization part is also a valuable addition. With the system being able to store data with timestamps, it is now possible to create growth graphs and compare patterns between different sessions. This turns the system into a progress-monitoring platform, rather than a simple measurement tool. Such visual feedback may be encouraging and informative especially when it is used to compare the current measurements with previous data, especially to users who are involved in strength training.

Meanwhile, there are also limitations that are presented in the results. It is also sensitive to the lighting conditions and the camera angle, typical of computer vision-based applications. When the subject is dimly lit, or the camera is not properly positioned, landmark detection can be compromised. Similarly, proper posture is a requisite to quality output. Even though most of these problems are addressed by the validation module, the system still requires the user to adhere to the protocols of capture appropriately. Another weakness is the inaccuracy in pseudo-3D modeling.

The elliptical model is efficient and helpful, yet it lacks complete anatomy of the real muscle forms. Hence, the values of the areas estimated need to be treated as approximations of anatomical values and not actual measurements. This constraint is reasonable in the context of the project since the objective is to track relative growth and not medical diagnosis. In general, the system has a good balance between practicality, efficiency, and accuracy. It is not intended to substitute medical-grade body composition analysis, but offers a sound and convenient system of fitness tracking and progress monitoring. These findings confirm that the multi-view landmark-driven analysis can be used as a promising method of longitudinal muscle hypertrophy monitoring.

5.5 Result Visualization

In the final report, the real visual evidence produced in the course of testing would be included in this section. The visual outputs are significant as they give direct support to the numerical results and simplify the interpretation of the system behavior. They should consist of a graph with the arm growth over time, a graph with the thigh growth over time, a screenshot of the system user interface, and a screenshot of the pose detection output. These images will reinforce the demonstration of the findings by demonstrating how the system works in real life and how the data stored is displayed in graphical format.



Figure 5. 4 Analysis Graph

6 Conclusions and Future Directions

Conclusion and Future Scope

Preamble to the Chapter

This chapter shows the conclusions of the work of the suggested system, overviews the major contributions, as well as identifies the limitations and potential future developments. It is the final part of the project, summarising the insights earned during the design, implementation, and evaluation processes. The chapter shows how the proposed AI-based system overcomes the challenges mentioned in the previous chapters and proves its efficiency as a feasible tool to monitor muscle hypertrophy. Moreover, it also talks about the areas where the system could be further enhanced and expanded, and it will serve as the roadmap to future research and development.

6.1 Conclusion

The project entitled AI-Based Multi-View Landmark-Driven Muscle Hypertrophy Tracking System with Longitudinal Growth Modelling exhibits a success of applying the computer vision and artificial intelligence methods to the field of fitness tracking and digital anthropometry. The system was made to address the drawbacks of conventional muscle measuring systems that tend to be manual, inconsistent in measurements, and subjective. Among the main successes of the system, there is the creation of an automated measurement framework. With the help of pose estimation, namely MediaPipe Pose, the system can identify anatomical landmarks in real-time and calculate useful body measurements. This has removed the use of manual tools like measuring tapes and minimized the human error, thus giving more consistent and repeatable results. Geometric calculations yielding important measurements of arm length, shoulder width, and thigh length are successfully obtained with the system with detected landmarks. The initial measurements in pixel units are then translated to real-world measurements through a scaling mechanism that relies on the user height. This guarantees that the findings are significant, and relevant to other sessions. One of the major innovations that have been brought in this project is the multi-view measurement approach. The system

can estimate width and thickness by getting both front and side views, which can be used to estimate the three-dimensional body structure. This method compensates the significant shortcoming of single-view systems that are unable to provide the view with depth data. Expanding on this, the system uses a pseudo-3D geometric modelling method, in which the cross-sections of muscles are modelled by an elliptical model. This allows the size of muscles to be estimated with regard to cross-sectional area giving a more realistic view of hypertrophy when compared to simple linear measures. Though this aids in growth tracking with time, this is an approximation, but it works well. The other notable aspect of the system is the pose validation mechanism, which only correct and consistent inputs are taken care of by the system. The system will greatly enhance reliability in measurements by rejecting improper postures, invalid captures and will minimize the noise in the data. The experimental assessment shows that the system gives a consistent measure at a tolerable error level of about 23 percent compared to a manual measure. This degree of accuracy is acceptable to fitness tracking and progress monitoring, where the relative changes are more significant than absolute accuracy. In general, the suggested system is a feasible, scalable, and cost-effective muscle hypertrophy monitoring system. It has managed to incorporate computer vision, geometric modelling, and data analysis into a cohesive system, and it can be applied practically to real-life contexts with the help of readily available hardware (e.g. standard webcams).

6.2 Key Contributions

The project presents a number of valuable contributions to the area of body measurement and fitness tracking using AI. The initial significant contribution is the design of a landmark-based measurement system. The system is able to compute measurements using pose estimation techniques to extract anatomical key points and use them as a starting point. This method implies uniformity and the absence of reliance on manual methods of measurements. The second contribution is that it implements a multi-view geometric modelling approach. This system combines front and side views unlike the traditional system which uses a single view to measure the width and thickness. This goes a long way to enhance the precision and accurateness of body measurements. The other important contribution is the introduction of a pseudo-3D muscle estimation technique.

The system gives the muscle size a meaningful representation by approximating muscle cross-sections as ellipses. This would enable improved monitoring of hypertrophy because the variation in both the dimensions is recorded, as opposed to the linear value. A validation-based measurement framework is also provided in the project. With pose validation checks, the system is able to process only correct and consistent inputs. This enhances dependability and minimizes errors due to poor posture or conditions of capture.

6.3 Limitations of the System

Although effective, the proposed system has some limitations that have to be taken into account. Among the main limitations is the fact that it is necessary to have the correct posture of the user. The system is based on visual input and detection of landmarks, so inaccurate measurements can happen because of wrong posture. Even though this can be addressed by the validation module, the dependency on user behavior cannot be fully overcome. The sensitivity to environmental factors, including lighting and camera angle is also another weakness. The lighting state or wrong positions of the camera can change the precision of pose estimating which in turn results to errors in deriving measurements. This is one of the challenges in computer vision systems which run under uncontrolled environments. The system is also based on the approximate geometric modeling. Pseudo-3D elliptical model is a simplified model of muscle structure, which is not entirely representative of human anatomy. Consequently, the size of the estimated muscle is an approximation of actual muscle size; this is not a measurement. Also, it is currently only implemented in single-user tracking. It is a single person system which did not support multiple users at a given time. This limits its use in situations like the gym or group training. These constraints can be used to identify areas where additional improvements can be done to improve the accuracy, robustness, and scalability of the system.

6.4 Future Scope

One of the major future enhancements of the proposed system is the integration of advanced 3D modeling techniques. Currently, the system estimates body measurements using pseudo-3D reconstruction from 2D images captured by a single camera. Although this approach provides satisfactory results and is cost-effective, it has certain limitations because it does not capture actual depth information. Factors such as camera angle, lighting conditions, body posture, and image quality can affect the accuracy of the measurements. To overcome these limitations, future versions of the system can incorporate advanced 3D modeling methods that generate a more realistic and accurate representation of the human body.

This enhancement can be achieved through the use of depth-sensing cameras, multi-camera systems, or AI-based 3D reconstruction algorithms. By capturing body information from multiple viewpoints or estimating depth more accurately, the system can create a complete three-dimensional body model rather than relying solely on 2D images. The generated 3D model would enable precise extraction of body measurements such as chest, waist, hip, arm, and leg dimensions, reducing approximation errors and improving overall measurement reliability.

Furthermore, advanced 3D modeling would allow the system to provide realistic body visualization and detailed fitness analysis. Users could view their body shape through a digital 3D avatar and compare changes over time to track fitness progress more effectively. The enhanced model could also support posture analysis, muscle growth monitoring, and personalized health recommendations. By integrating advanced 3D reconstruction techniques with artificial intelligence, the system would become more accurate, robust, and suitable for a wider range of applications, including fitness tracking, healthcare monitoring, and personalized body assessment.

6.5 Sustainable Development Goals (SDG) Alignment

The suggested system is in line with various United Nations Sustainable Development Goals. It also addresses SDG 3: Good Health and Well-being since it leads to physical fitness tracking and a healthier lifestyle among people. The system assists in making sound decisions in fitness and health management by offering a mechanism of monitoring muscle development. It also supports SDG 9: Industry, Innovation and Infrastructure by implementing artificial intelligence and computer vision to a real-life scenario. The system proves the way the emerging technologies can be adopted to address daily problems and be innovative. Also, the project endorses SDG 4: Quality Education too, as it illustrates how the AI concepts can be practically implemented. It is a learning tool to students and researchers within the realms of computer vision, data analysis as well as system design.

6.6 Final Summary

The presented system also shows the successful application of computer vision and artificial intelligence in addressing a real-life issue during the tracking of fitness. The system offers integrated pose estimation, multi-view geometry, mathematical model, and data visualization, making it a comprehensive platform of muscle hypertrophy tracking. It is capable of converting the subjective traditional techniques into a measureable and automated procedure with reliable and consistent outcomes. Although the system has some constraints, it offers a good base on which other research and development can be based. In general, the project points to the opportunities of AI-based solutions in improving individual health monitoring and shows that the available technologies can be used to create pragmatic and powerful systems.

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Summary of References Usage

These references collectively support:

- Pose estimation techniques (MediaPipe, OpenPose)
- Geometric modeling and measurement computation
- Digital anthropometry principles
- Real-time computer vision systems
- Mathematical modeling and data analysis